

Quantification of Structural Uncertainties in RANS Turbulence Models

by

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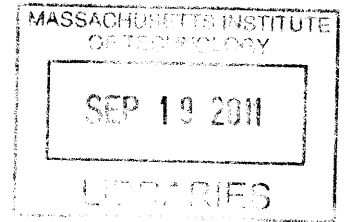
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Abstract

This thesis presents an approach for building a statistical model for the structural uncertainties in Reynolds averaged Navier-Stokes (RANS) turbulence models. This approach solves an inference problem by comparing the results of RANS calculations to direct numerical simulation. The adjoint method is used to efficiently solve an inverse problem to determine the RANS turbulent viscosity field that most accurately reproduces the mean flow field computed by direct numerical simulation. The discrepancy between the inferred turbulent viscosity and the turbulent viscosity predicted by RANS is modeled as a Gaussian random field. Finally, the uncertainty in the turbulent viscosity field is propagated to the quantities of interest. Results are first presented for turbulent flow through a straight channel. To model the uncertainty in more complex flows, the procedure is repeated for a collection of flows through randomly generated geometries.

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Chapter 1

Introduction

1.1 Motivation

In many engineering applications involving turbulent flows, resolving the effect of turbulence is critical to accurately estimating and optimizing the performance. If no turbulence model is used, resolving the effect of turbulence requires extremely fine meshes that capture the motions at the smallest dissipative scale, i.e. the Kolmogorov scale. This approach, i.e. relying on very fine meshes to resolve the small scale turbulent motions, is referred to as direct numerical simulation (DNS). The size of the computational mesh required to perform DNS grows rapidly with the Reynolds number of the flow [13]. Thus, directly computing the effect of turbulence is typically too expensive. To decrease the computational cost of simulating turbulent flows, a number of methods have been developed to model the effect of turbulence. Rather than directly resolve the fine scales of turbulent motion, these models introduce terms into the Navier-Stokes equations that model the effect of small scale turbulent motions on the mean flow field. Computational methods based on solving the Reynolds averaged Navier-Stokes (RANS) equations are currently the most popular choice for simulating flow problems that involve turbulence. Solving the RANS equations determines the statistically-averaged flow field without regard to the fine scale turbulent structures, thus eliminating the need for very fine meshes. RANS-based simulation is thus relatively inexpensive as compared to DNS. This reduction in computational cost

makes RANS ideal for use in the engineering design process, where the flow around numerous design iterations must be simulated during an optimization procedure.

The RANS equations are formulated by Reynolds averaging the Navier-Stokes equations. For incompressible flows, the velocity and pressure fields can be decomposed into a mean and a fluctuating component. This decomposition is referred to as the Reynolds decomposition. The RANS equations are obtained by inserting the Reynolds decomposition of the velocity and pressure fields into the Navier-Stokes equations and taking the Reynolds average. The RANS equations for the mean flow field are identical to the original Navier-Stokes equations, with an additional apparent stress involving the components of the fluctuating velocity, known as the Reynolds stress. The key difficulty is that the transport equation for the Reynolds stress involves higher order correlations, and the transport equation for the higher order correlations involves higher order correlations still. Thus, solving the RANS equations is a closure problem. RANS turbulence models are used to define closure relations that allow the RANS equations to be solved. One specific class of turbulence models, the Boussinesq turbulent viscosity models, relate the Reynolds stress tensor to the mean velocity field by prescribing a turbulent viscosity acting on the mean flow field [13]. Within the class of Boussinesq models, a variety of methods have been proposed for estimating the turbulent viscosity field. For simple flows, these models typically produce good estimates of the effect of turbulence. However, for more complex flows, the mean flow fields computed using turbulent viscosity models show significant discrepancies with experimental results. Turbulent viscosity models are especially inaccurate for flows that experience or are close to separation, or where the streamline curvature is large [14][18][15], flow conditions that are commonly observed in complex aerospace applications.

Turbulence models introduce uncertainty into the computation of the flow field. Since the value of the Reynolds stress is unknown, the discrepancy between the true flow field and the flow field computed using a turbulence model is also uncertain. This uncertainty is often referred to as model uncertainty or structural uncertainty, since it originates as a consequence of the assumptions made about the underlying

relation between the mean flow field and the Reynolds stress tensor. Since this form of uncertainty can theoretically be reduced (for example, by devising better models for estimating the turbulent viscosity), structural uncertainties represent an epistemic uncertainty. Estimating the uncertainty in current turbulence models is important for numerous reasons. If, for example, the relative performance of two competing designs is to be compared, it is important to know if the differences in the computed performance are large relative to the uncertainty in the computation. This can inform whether a more detailed simulation is required to provide a conclusive comparison.

1.1.1 Quantifying Structural Uncertainty

In this thesis, an approach for quantifying structural uncertainty in RANS simulations of complex flows is presented. This approach consists of three steps: an inverse modeling step, a statistical modeling step, and an uncertainty propagation step. The inverse modeling step generates data which is in turn used to construct a statistical model of structural uncertainties. The results of direct numerical simulation are used to determine the “true” RANS turbulent viscosity that most accurately reproduces the DNS flow field. To invert the turbulent viscosity field, the inverse problem is formulated as a constrained optimization problem, and the resulting optimization problem is solved using gradient based optimization techniques. For computational efficiency, the adjoint method is used to compute the sensitivity gradient. The true turbulent viscosity fields are stored together with the mean flow field and turbulence properties. The inverse modeling step reduces the problem of quantifying the sources of uncertainty to a statistical data analysis problem. In the statistical modeling step, the data generated in the inverse modeling step is used to construct a statistical model of the uncertainty in the calculated RANS turbulent viscosity field. The level of uncertainty in the turbulent viscosity field is correlated to various geometric and flow features, allowing the statistical model of uncertainty to be applied to any RANS flow solution. Finally, this statistical model is sampled to propagate the uncertainty in the RANS turbulent viscosity field to the quantities of interest.

The key assumption made in formulating this approach is that the uncertainty

in RANS computations can be largely attributed to the inability of current RANS models to estimate the true turbulent viscosity. This assumption is validated by considering the results of the inverse modeling step, and motivates the approach of characterizing the discrepancy between the computed RANS turbulent viscosity and the true turbulent viscosity fields. These discrepancies can be viewed as a result of uncertainty in the estimation of the turbulent viscosity field, and reflect the inability of the Reynolds stress tensor to be approximated accurately by solving for a small number of transport scalars, e.g. the turbulence kinetic energy and specific dissipation rate in the $k - \omega$ model.

This thesis is organized as follows. Chapter 2 describes the use of the adjoint method for solving inverse problems in which a set of model parameters is estimated, and derives the adjoint system for the mean flow equations. Chapter 3 presents the results for quantifying the structural uncertainty in turbulent flow through a straight channel. Chapter 4 applies the same framework to quantify the structural uncertainty in more general flows. Conclusions and discussion of future work is presented in chapter 5.

1.2 Previous Research

Due to the widespread use of RANS turbulence models in industry, attempts have been made to quantify the structural uncertainties in RANS simulations. The work of Platteeuw et al. [12] uses a collection of experimental results and direct numerical simulations to determine the distributions of the closure coefficients of the $k - \epsilon$ model for turbulent flow over a flat plate. The uncertainty in the closure coefficients is then propagated to estimate the uncertainty in the friction coefficient using the Probabilistic Collocation Method. The predicted level of uncertainty in the friction coefficient appears reasonable as the experimental data falls within the 99% confidence intervals around the mean friction coefficient profile. The focus of this work is on developing an efficient method of propagating uncertainty rather than the characterization of the sources of uncertainty. For example, the assumed distributions of some parameters

must be guessed, due to a lack of available experimental or direct numerical simulation data. Also, the authors have only applied their approach to estimate the uncertainty in quantities of interest for turbulent flow over a flat plate, a relatively simple test case. Their approach will likely have difficulty predicting the uncertainty in more complex flows where the uncertainty in quantities of interest cannot be accurately captured by estimating uncertainty in a small number of model parameters.

The recent work of Cheung et al. [2] applies a Bayesian uncertainty analysis framework to estimate the uncertainty in RANS simulations of turbulent flow over a flat plate. The authors consider three flow cases distinguished by a favorable, zero, and adverse pressure gradient for which experimental data is available. The closure coefficients of the the Spalart-Allmaras RANS model are treated as random variables and the probability distributions of the closure coefficients are estimated by solving a Bayesian inverse problem. The Bayesian framework incorporates experimental calibration data which is assumed to have some level of uncertainty with prior estimates of the uncertainty in the model parameters. The prior distributions of the closure coefficients are taken to be uniform distributions over the plausible range of values for each coefficient. Once the posterior distributions of the closure coefficients have been computed, the uncertainty is propagated to estimate the uncertainty in the shear stress at a particular location within the domain. Although the Bayesian uncertainty analysis framework is a popular choice for estimating uncertainty, there are some shortcomings that limit its applicability to estimating the uncertainty in RANS simulations. The Bayesian framework is typically only applicable for problems where only a small number of uncertain parameters need to be estimated. For complex flows where the spatial variation in uncertainty can be large, the uncertainty can not be captured by only considering a small number of model parameters. Also, the Bayesian framework requires prior information on the model parameters. The posterior estimate of uncertainty is sensitive to the choice of prior, and an informative prior may not always be available.

Other recent work has sought to use DNS to tune existing RANS turbulence models. Since DNS resolves all of the relevant scales of turbulent motion, the results

are extremely high fidelity, and have thus been used to determine the accuracy of turbulence models. Some recent examples include the work of Venayagamoorthy et. al., where the results of direct numerical simulation are used to develop trends for the various tuning parameters of the $k - \epsilon$ model for stratified flows [16]. They note that the DNS results do not always present clear trends, and that it may be up to the modeler to choose the trend they feel most appropriate. Kim et al. provide a detailed comparison between the results of DNS with a variety of RANS models for turbulent mixed convection [7]. They conclude that some models are superior in capturing the effects of buoyancy, and that the performance of these models is highly sensitive to the choice of tuning parameters. Comparisons like these shed significant light on the uncertainties in RANS models, but typically must be performed on a case by case basis.

Chapter 2

The adjoint method for inverse problems

2.1 Introduction

In this chapter, the use of the adjoint method for solving inverse problems is described. In section 2.2, the procedure for recasting inverse problems as optimization problems is outlined. Section 2.3 provides an abstract formulation of the adjoint equations for solving the resulting optimization problem. The continuous adjoint system and sensitivity gradient are also derived for the flows of interest.

2.2 Formulating inverse problems as optimization problems

In solving inverse problems, the goal is to determine the set of model parameters m that yields the closest agreement between the output of the system and the observables d . The systems of interest in this work are governed by some PDE, so the objective is to determine the set of parameters such that

$$d = G(m),$$

where $G(m)$ represents the evaluation of the PDE with the model parameters m . In order to apply the adjoint method to this class of problems, the inverse problem is first cast as an optimization problem. In this optimization problem, the objective function is chosen to measure the difference between the observables and the output of the model evaluated for some choice of model parameters $\tilde{m}$, i.e.

$$\mathbf{J}(\tilde{m}) = \|d - G(\tilde{m})\|_2^2.$$

When the norm of the difference between the output and the observables is zero, the two must agree, and the inverse problem has been solved. The solution to the inverse problem is then defined as

$$m = \underset{\tilde{m}}{\operatorname{argmin}} \mathbf{J}(\tilde{m}). \quad (2.1)$$

An advantage to this approach is the handling of any constraints specified for the model parameters. These constraints are simply adopted as constraints in the optimization problem specified by equation 2.1.

A simple procedure to compute the optimal solution to 2.1 is to first compute a descent direction $\partial \mathbf{J} / \partial m$, which represents the sensitivity gradient of the objective function with respect to the model parameters. When the step size λ is small, the solution can be updated by setting

$$m^{k+1} = m^k - \lambda \frac{\partial \mathbf{J}}{\partial m}.$$

To first order

$$\mathbf{J} + \delta \mathbf{J} = \mathbf{J} + \frac{\partial \mathbf{J}^T}{\partial m} \delta m = \mathbf{J} - \lambda \frac{\partial \mathbf{J}^T}{\partial m} \frac{\partial \mathbf{J}}{\partial m},$$

and thus there exists some λ such that objective function value is decreased by updating in this fashion [5].

The key difficulty in this approach is evaluating the sensitivity gradient. One method of approximating the sensitivity gradient is to evaluate the objective function by adding a small variation δm_i in each of the model parameters and approximating

the sensitivity gradient as

$$\frac{\partial \mathbf{J}}{\partial m_i} \approx \frac{\mathbf{J}(m_i + \delta m_i) - \mathbf{J}(m_i)}{\delta m_i}$$

Computing the sensitivity gradient in this manner requires $\dim(m) + 1$ evaluations of the objective function. In this work, evaluating the objective function requires solving a PDE, and $\dim(m)$ is generally large. Thus, computing the sensitivity gradient in this manner is very computationally expensive. This difficulty motivates the use of the adjoint method for computing the sensitivity gradient.

2.3 Mathematical formulation of the adjoint equations

As mentioned previously, the systems of interest are governed by PDEs, namely the RANS equations. The objective function is then a function of both the model parameters and the solution to the PDE evaluated with the model parameters, referred to as $u(m)$. In this case,

$$\mathbf{J} = \mathbf{J}(u(m), m),$$

and a change in the model parameters results in a change in the objective function value

$$\delta \mathbf{J} = \left[\frac{\partial \mathbf{J}}{\partial u} \right]^T \delta u + \left[\frac{\partial \mathbf{J}}{\partial m} \right]^T \delta m, \quad (2.2)$$

where $\delta \mathbf{J}$, δu , and δm are infinitesimal. It is assumed that the governing equation for the PDE that controls the system can be written as

$$R(u, m) = 0.$$

Linearizing the governing equation, the variation δR can be written as

$$\delta R = \left[\frac{\partial R}{\partial u} \right] \delta u + \left[\frac{\partial R}{\partial m} \right] \delta m = 0.$$

This variation is zero, so the linearized governing equation can be multiplied by a costate ψ and introduce the linearized equation as a “constraint” in the minimization problem. Equation 2.2 can thus be replaced by

$$\begin{aligned}\delta\mathbf{J} &= \frac{\partial\mathbf{J}^T}{\partial u} \delta u + \frac{\partial\mathbf{J}^T}{\partial m} \delta m - \psi^T \left(\left[\frac{\partial R}{\partial u} \right] \delta u + \left[\frac{\partial R}{\partial m} \right] \delta m \right) \\ &= \left(\frac{\partial\mathbf{J}^T}{\partial u} - \psi^T \left[\frac{\partial R}{\partial u} \right] \right) \delta u + \left(\frac{\partial\mathbf{J}^T}{\partial m} - \psi^T \left[\frac{\partial R}{\partial m} \right] \right) \delta m\end{aligned}$$

To eliminate the direct dependence of the objective function on the solution u , the costate ψ is chosen to satisfy the adjoint equation

$$\left[\frac{\partial R}{\partial u} \right]^T \psi = \frac{\partial\mathbf{J}}{\partial u}. \quad (2.3)$$

If ψ satisfies 2.3, which in this case is a linear PDE, the variation in the objective function becomes

$$\delta\mathbf{J} = \mathcal{G} \delta m$$

where the sensitivity gradient $\mathcal{G}$ is defined as

$$\mathcal{G} = \frac{\partial\mathbf{J}^T}{\partial m} - \psi^T \left[\frac{\partial R}{\partial m} \right].$$

The sensitivity gradient can be computed by solving the original PDE once, followed by one additional solve of the adjoint equation. The computational cost of solving is roughly the same as the cost of solving the original PDE. Thus, the sensitivity gradient with respect to all of the model parameters can be computed at roughly twice the cost of solving the original PDE.

2.3.1 Adjoint system for RANS flow in a straight channel

The inverse problem of interest in this work is to determine the turbulent viscosity field that produces a RANS flow solution that is closest to the flow solution predicted by direct numerical simulation. In this problem, the model parameters that need to

be inverted are the values of a continuous field. Numerically, the turbulent viscosity field must be discretized, and for complex flows on arbitrary domains, the dimension of the resulting discretization will be quite large. Thus, this inverse problem is well suited to applying the adjoint approach.

To cast this inverse problem as an optimization problem, an objective function must be formed that measures the difference between the RANS flow velocity $u(\nu_T)$ computed with a specified turbulent viscosity and the DNS flow velocity u_{DNS} . The objective function is chosen as

$$\mathbf{J}(u(\nu_T)) = \|u(\nu_T) - u_{DNS}\|_{L^2}^2$$

For physical reasons, the turbulent viscosity is required to be non-negative, so the minimization statement is given as

$$\begin{aligned} \min \quad & \|u(\nu_T) - u_{DNS}\|_{L^2}^2 \\ \text{s.t.} \quad & \nu_T \geq 0 \end{aligned} \tag{2.4}$$

The adjoint system corresponding to this objective function for steady turbulent flow in a straight walled channel can now be derived. The domain of interest extends from the channel wall at $y/\delta = 0$ to the channel center at $y/\delta = 1$ where δ is the channel half-width. For steady incompressible turbulent flow in a periodic straight-walled channel, the mean flow equations with normalized density are

$$-\frac{d}{dy} \left(\nu_{\text{eff}} \frac{du}{dy} \right) = f, \tag{2.5}$$

$$u(0) = 0 \quad \frac{du}{dy}(1) = 0$$

where u is the mean axial flow velocity, $\nu_{\text{eff}} = \nu_T + \nu$ is the effective viscosity, and f is a constant forcing applied to drive the flow (e.g. a uniform pressure gradient). To determine the corresponding adjoint equations, the tangent set of equations is first formed by substituting $u = \bar{u} + \delta u$ and $\nu_T = \bar{\nu}_T + \delta \nu_T$:

$$\frac{d}{dy} \left((\bar{\nu}_T + \nu) \frac{d\delta u}{dy} + \delta \nu_T \frac{d\bar{u}}{dy} \right) = 0$$

$$\delta u(0) = 0 \quad \frac{d\delta u}{dy}(1) = 0.$$

For the rest of the derivation, the overbar notation is omitted and it is assumed that the system is linearized about the states u and ν_T . The linearized objective function is

$$\delta \mathbf{J} = \int_0^1 2(u - u_{DNS}) \delta u \, dy. \quad (2.6)$$

Introducing the adjoint velocity $\hat{u}$, equation (2.10) can be rewritten as

$$\delta \mathbf{J} = \int_0^1 2(u - u_{DNS}) \delta u \, dy + \int_0^1 \hat{u} \frac{d}{dy} \left((\nu_T + \nu) \frac{d\delta u}{dy} + \delta \nu \frac{du}{dy} \right) dy.$$

Integration by parts gives

$$\begin{aligned} \delta \mathbf{J} = & \int_0^1 2(u - u_{DNS}) \delta u \, dy + \int_0^1 \delta u \frac{d}{dy} \left((\nu_T + \nu) \frac{d\hat{u}}{dy} \right) dy - \int_0^1 \delta \nu_T \frac{du}{dy} \frac{d\hat{u}}{dy} dy \\ & + \hat{u} \left((\nu_T + \nu) \frac{d\delta u}{dy} + \delta \nu_T \frac{du}{dy} \right) \Big|_0^1 - (\nu_T + \nu) \frac{d\hat{u}}{dy} \delta u \Big|_0^1. \end{aligned}$$

All terms involving δu are set to zero, arriving at the adjoint equation

$$\frac{d}{dy} \left((\nu_T + \nu) \frac{d\hat{u}}{dy} \right) = -2(u - u_{DNS}), \quad (2.7)$$

with corresponding boundary conditions

$$\hat{u}(0) = 0 \quad \frac{d\hat{u}}{dy}(1) = 0.$$

The sensitivity of the objective function to the turbulent viscosity can be computed as

$$\frac{\partial \mathbf{J}}{\partial \nu_T} = -\frac{d\hat{u}}{dy} \frac{du}{dy}. \quad (2.8)$$

The adjoint system given by equation 2.7 is qualitatively identical to the primal system given by equation 2.5. Thus, if an efficient method to solve the primal equation is available, the adjoint system can be solved in a similar manner.

2.3.2 Regularization

Due to the Neumann boundary condition applied to the adjoint equation, the sensitivity gradient is identically zero at the channel center. This implies that the inverse problem solved by 2.4 is ill-posed. Physically, this arises because the velocity gradient is zero at the channel center. When the velocity gradient is zero at some location, changing the turbulent viscosity at this location will not change the flow solution, and thus will not change the objective function value. Since a unique turbulent viscosity field must be determined, additional information is introduced to regularize the solution. Specifically, the total variation of the turbulent viscosity field is penalized by introducing an additional term to the objective function. The new objective function is

$$\mathbf{J}(u(\nu_T), \nu_T) = \|u(\nu_T) - u_{DNS}\|_{L^2}^2 + \varepsilon \|\nabla \nu_T\|_{L^2}^2$$

where the regularization parameter ε is chosen to be small relative to the channel width. In practice, the regularization parameter must be chosen carefully to ensure that the optimal solution is not overly smoothed. Since the regularization term in the objective function does not involve the flow solution $u(\nu_T)$, it does not need to be included in the derivation of the adjoint equations. Instead, the sensitivity gradient of this term with respect to the turbulent viscosity can simply be added to the adjoint sensitivity gradient computed using 2.8.

2.3.3 Adjoint system for the mean flow equations

In this section, the adjoint sensitivity gradient used for solving the RANS inverse problem for arbitrary flows is derived. The derivation proceeds in much the same way as for flow in a straight channel. Although only steady flows are considered in this thesis, the adjoint system for the unsteady problem is derived. This is motivated by the fact that the solver used in this work computes the adjoint solution to the steady problem by computing the steady state solution of the unsteady adjoint equations. The mean flow equations take the form

$$\frac{\partial u}{\partial t} + u \cdot \nabla u + \nabla p - \nabla \cdot (\nu_{\text{eff}} \nabla u) - f = 0,$$

$$\nabla \cdot u = 0$$

in the spatial domain Ω in a time interval $[0, T]$. In this work, only flows with Dirichlet boundary conditions on the entire boundary $\partial\Omega$ are considered, i.e.

$$u = 0, \quad x \in \partial\Omega.$$

Also, the boundaries are assumed to be solid walls, so the zero flux condition

$$u \cdot \hat{n} = 0, \quad x \in \partial\Omega.$$

is enforced at the boundaries. The turbulent viscosity must also be zero at the solid boundaries:

$$\nu_T = 0, \quad x \in \partial\Omega.$$

Linearizing the mean flow equations about the flow solution $u(\nu_T)$, the linearized governed equation is

$$\frac{\partial \delta u}{\partial t} + \mathcal{L}_{(u, \nu_T)} + \nabla \delta p = 0$$

$$\nabla \cdot \delta u = 0$$

$$\delta u = 0, \quad x \in \partial\Omega$$

$$\delta u \cdot \hat{n} = 0, \quad x \in \partial\Omega$$

$$\delta \nu_T = 0, \quad x \in \partial\Omega$$

where the linearized operator $\mathcal{L}_{(u, \nu_T)}$ is defined as

$$\mathcal{L}_{(u, \nu_T)} = u \cdot \nabla \delta u + \delta u \cdot \nabla u - \nabla \cdot ((\nu + \delta \nu_T) \nabla u) - \nabla \cdot (\nu_{\text{eff}} \nabla \delta u) \quad (2.9)$$

The objective function of interest is essentially the same as that described in section 2.3.1, except that now all components of the velocities are considered rather than just the axial velocity. The objective function must also be integrated in time in order to derive the unsteady adjoint equations. The linearized objective function is then given by

$$\delta \mathbf{J} = \int_0^T \iiint_{\Omega} 2(u - u_{DNS}) \cdot \delta u \, dx \, dt. \quad (2.10)$$

Introducing the adjoint variables $\hat{u}$ and $\hat{p}$, and combining the linearized objective function and mean flow equations:

$$\begin{aligned} \delta \mathbf{J} &= \int_0^T \iiint_{\Omega} 2(u - u_{DNS}) \cdot \delta u \, dx \, dt \\ &+ \int_0^T \iiint_{\Omega} \frac{\partial \delta u}{\partial t} \cdot \hat{u} + \mathcal{L}_{(u, \nu_T)} \cdot \hat{u} + \hat{p}(\nabla \cdot \delta u) \, dx \, dt \end{aligned} \quad (2.11)$$

Integrating the time derivative by parts,

$$\int_0^T \hat{u} \cdot \frac{\partial \delta u}{\partial t} \, dt = \hat{u} \cdot \delta u|_0^T - \int_0^T \delta u \cdot \frac{\partial \hat{u}}{\partial t} \, dt \quad (2.12)$$

The remaining terms are integrated by parts in space, and the appropriate boundary conditions are enforced on u and δu :

$$\begin{aligned} \iiint_{\Omega} \hat{p}(\nabla \cdot \delta u) \, dx &= \iint_{\partial\Omega} \hat{p} \, \delta u \cdot \vec{n} \, ds - \iiint_{\Omega} \delta u \cdot \nabla \hat{p} \, dx \\ &= - \int_0^T \iiint_{\Omega} \delta u \cdot \nabla \hat{p} \, dx \end{aligned}$$

$$\iiint_{\Omega} \hat{u} \cdot \nabla \delta p \, dx = \iint_{\partial\Omega} \delta p \, \hat{u} \cdot \vec{n} \, ds - \iiint_{\Omega} \delta p \nabla \cdot \hat{u} \, dx$$

$$\begin{aligned} \iiint_{\Omega} (u \cdot \nabla \delta u) \cdot \hat{u} \, dx &= \iint_{\partial\Omega} (\delta u \cdot \hat{u}) u \cdot \vec{n} \, ds \\ &\quad - \iiint_{\Omega} \delta u \cdot (u \cdot \nabla \hat{u}) + (\nabla \cdot u)(\delta u \cdot \hat{u}) \, dx \\ &= - \iiint_{\Omega} \delta u \cdot (u \cdot \nabla \hat{u}) \, dx \end{aligned}$$

$$\begin{aligned} \iiint_{\Omega} (\nabla \cdot (\nu_{\text{eff}} \nabla \delta u)) \cdot \hat{u} \, dx &= \iint_{\partial\Omega} \nu_{\text{eff}} \, \hat{u} \cdot (\nabla \delta u \cdot \vec{n}) - \nu_{\text{eff}} (\delta u \cdot (\nabla \hat{u})) \cdot \vec{n} \, ds \\ &\quad + \iiint_{\Omega} \delta u \cdot (\nabla \cdot (\nu_{\text{eff}} \nabla \hat{u})) \, dx \\ &= \iint_{\partial\Omega} \nu_{\text{eff}} \, \hat{u} \cdot (\nabla \delta u \cdot \vec{n}) \, ds \\ &\quad + \iiint_{\Omega} \delta u \cdot (\nabla \cdot (\nu_{\text{eff}} \nabla \hat{u})) \, dx \end{aligned}$$

$$\begin{aligned} \iiint_{\Omega} (\nabla \cdot (\delta \nu_T \nabla u)) \cdot \hat{u} \, dx &= \iint_{\partial\Omega} \delta \nu_T ((\nabla u) \cdot \hat{u}) \cdot \vec{n} \, ds - \iiint_{\Omega} \delta \nu_T \nabla u : \nabla \hat{u} \, dx \\ &= - \iiint_{\Omega} \delta \nu_T \nabla u : \nabla \hat{u} \, dx \end{aligned}$$

Combining these relations,

$$\begin{aligned} \delta \mathbf{J} &= \int_0^T \iiint_{\Omega} 2(u - u_{DNS}) \cdot \delta u \, dx \, dt \\ &= \int_0^T \iiint_{\Omega} \delta u \cdot \left(-\frac{\partial \hat{u}}{\partial t} - u \cdot \nabla \hat{u} + \nabla u \cdot \hat{u} - \nabla \cdot (\nu_{\text{eff}} \nabla \hat{u}) + \nabla \hat{p} \right) - \delta p \nabla \cdot \hat{u} \, dx \, dt \\ &\quad + \iiint_{\Omega} \hat{u}(T) \cdot \delta u(T) - \hat{u}(0) \cdot \delta u(0) \, dx \\ &\quad + \int_0^T \iint_{\partial\Omega} \delta p \, \hat{u} \cdot \vec{n} - \nu_{\text{eff}} ((\nabla \delta u) \cdot \hat{u}) \cdot \vec{n} \, ds \, dt \\ &\quad + \int_0^T \iiint_{\Omega} \delta \nu_T \nabla u : \nabla \hat{u} \, dx \, dt \end{aligned}$$

To determine the adjoint sensitivity gradient with respect to the turbulent viscosity, all terms involving δu and δp are made to vanish by choosing the adjoint variables to satisfy the continuous adjoint equations:

$$-\frac{\partial \hat{u}}{\partial t} - u \cdot \nabla \hat{u} + \nabla u \cdot \hat{u} - \nabla \cdot (\nu_{\text{eff}} \nabla \hat{u}) + \nabla \hat{p} = 2(u - u_{DNS}) \quad (2.13)$$

$$\nabla \cdot \hat{u} = 0. \quad (2.14)$$

The corresponding adjoint boundary conditions are

$$\hat{u} \cdot \vec{n} = 0, \quad x \in \partial\Omega \quad (2.15)$$

$$\hat{u} = 0, \quad x \in \partial\Omega.$$

Since the steady state solution of the adjoint equation is computed, the choice of terminal condition is unimportant. For simplicity, the terminal condition $\hat{u}(T) = 0$ is applied for the adjoint velocity. The adjoint sensitivity gradient is computed as

$$\frac{\partial \mathbf{J}}{\partial \nu_T} = \nabla u : \nabla \hat{u}. \quad (2.16)$$

Since a terminal condition is specified, the adjoint equation 2.13 must be solved backward in time. In practice, one can compute the adjoint solution forward in time by substituting $\tau = T - t$. The resulting adjoint equation is then

$$\frac{\partial \hat{u}}{\partial \tau} - u \cdot \nabla \hat{u} + \nabla u \cdot \hat{u} - \nabla \cdot (\nu_{\text{eff}} \nabla \hat{u}) + \nabla \hat{p} = 2(u - u_{DNS}) \quad (2.17)$$

Equation 2.17 is very similar in form to the original mean flow equations. The adjoint variable is convected by the mean flow, diffuses with the same effective viscosity, and is driven by the gradient in the adjoint pressure variable. The biggest differences are that the adjoint equations are linear, and that new forcing terms arise in the adjoint equations.

For the inverse problem of interest, the sensitivity gradient computed by equa-

tion 2.16 can lead to an ill-posed problem. If the velocity gradient tensor is identically zero somewhere in the flow, the objective function value is insensitive to changing the turbulent viscosity at this location, and the inverse problem is ill-posed. This issue is again remedied by introducing the same regularization described in the previous section. The contribution to the sensitivity gradient due to the regularization term is computed independently of the adjoint sensitivity gradient, and the two are added together when performing the optimization.

Chapter 3

Quantifying turbulence model uncertainty for flow through a straight channel

3.1 Introduction

In this chapter, the approach described in chapter 1 is applied to quantify the model uncertainty in turbulent flow through a periodic straight walled channel. This relatively simple test case was chosen to validate the framework and develop strategies for solving the RANS inverse problem and constructing statistical models of the structural uncertainties.

3.2 Numerical computation of the adjoint sensitivity gradient

This section derives the adjoint sensitivity gradient for flow through a straight walled channel. The domain of interest extends from the channel wall, corresponding to $y/\delta = 0$, to the channel center line at $y/\delta = 1$. The initial turbulent viscosity profile is computed using the Willcox $k - \omega$ turbulent model. The finite difference method

is used to solve the equations governing momentum and the transport of turbulence kinetic energy and specific dissipation rate:

$$-\frac{d}{dy} \left(\left(\nu + \frac{k}{\omega} \right) \frac{du}{dy} \right) = f, \quad (3.1)$$

$$-\frac{d}{dy} \left(\left(\nu + \sigma^* \frac{k}{\omega} \right) \frac{dk}{dy} \right) = \beta^* k \omega, \quad (3.2)$$

$$-\frac{d}{dy} \left(\left(\nu + \sigma \frac{k}{\omega} \right) \frac{d\omega}{dy} \right) = \beta \omega^2, \quad (3.3)$$

with model closure coefficients

$$\beta = 3/40, \quad \beta^* = 9/100, \quad \sigma = 1/2, \quad \sigma^* = 1/2.$$

The forcing f is chosen to be unity everywhere in the domain. Solving equations 3.1-3.3 provides the initial estimate for the turbulent viscosity profile that will be optimized using the adjoint sensitivity gradient.

To compute the sensitivity gradient, the adjoint equation derived in chapter 2 is first solved.

$$\frac{d}{dy} \left((\nu_T + \nu) \frac{d\hat{u}}{dy} \right) = -2(u - u_{DNS}). \quad (3.4)$$

The right hand side of this equation involves the velocity profile computed using direct numerical simulation. The DNS flow profile used in this work is taken from a database provided by Moser, Kim, and Mansour [9]. This database contains DNS results computed for flow through a straight channel at the friction Reynolds numbers of approximately $Re_\tau = 180, 395, \text{ and } 590$, where the friction Reynolds number is defined as

$$Re_\tau = \frac{u_\tau \delta}{\nu}, \quad u_\tau = \sqrt{\tau_w / \rho}.$$

The velocity profiles in this database are the time-averaged profiles computed using direct numerical simulation. A comparison between the RANS and DNS velocity profiles is shown in figure 4-7. Clearly, the $k-\omega$ model tends to overestimate the level of turbulent dissipation, and the resulting velocity magnitude is smaller everywhere

in the domain.

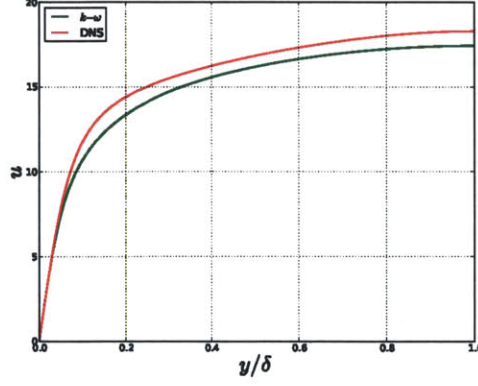


Figure 3-1: Comparison of RANS and DNS velocity profiles for $Re_\tau = 180$.

Since equation 3.4 is linear and elliptic, a natural solution approach is the finite element method. Equation 3.4 is discretized using linear finite elements, and the proper Dirichlet and Neumann boundary conditions are imposed at the domain boundaries. The adjoint solution $\hat{u}$ can then be used to compute the adjoint sensitivity gradient according to equation 2.8. The initial adjoint solution and sensitivity gradient are shown in figure 3-2. The adjoint solution can be interpreted as the change in the objective function value per unit change in the the RANS mean velocity at a given location. Since the magnitude of the RANS velocity predicted by the $k - \omega$ model is smaller than the DNS velocity everywhere in the domain, it is expected that increasing the RANS velocity will decrease the objective function value. This agrees with the plot of the adjoint solution.

The adjoint sensitivity gradient of the turbulent viscosity field represents the change in the objective function value per unit change in the turbulent viscosity at a given location. The sensitivity gradient shown in figure 3-2 agrees with intuition. Changing the turbulent viscosity near the wall, where the velocity gradient is largest, will have the largest global impact on the RANS mean velocity, and thereby has the largest impact on the objective function value. Increasing the turbulent viscosity at the wall will decrease the velocity magnitude globally, thereby increasing the objective function value. Thus, the initial adjoint sensitivity gradient is positive at the wall. The sensitivity gradient at the channel center ($y/\delta = 1$) is zero. The objective

function value is completely insensitive to changes in the turbulent viscosity at this location.

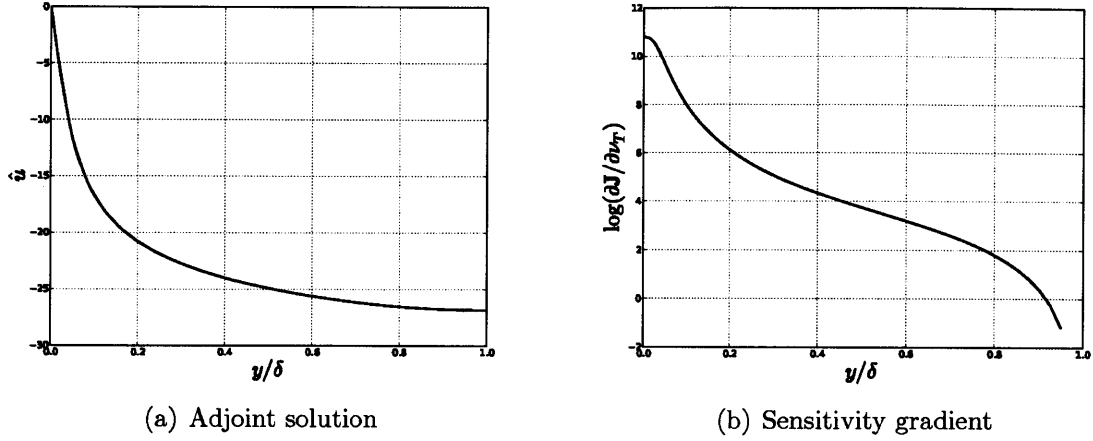


Figure 3-2: Initial adjoint solution and log-sensitivity gradient

3.3 Optimization procedure

The sensitivity gradient depicted in figure 3-2 represents a descent direction for the optimization problem of determining the true turbulent viscosity profile. As the turbulent viscosity is updated and the resulting velocity field changes, the sensitivity gradient is recomputed by solving the adjoint equation. The turbulent viscosity and velocity profiles are updated iteratively until the velocity field converges. The convergence of the velocity field is measured by considering the objective function value. This value will cease to change once the velocity field computed with the updated turbulent viscosity profile no longer changes.

The optimization problem described in chapter 2 had a single inequality constraint, namely that the turbulent viscosity field must remain non-negative. Physically, this corresponds to the requirement that the turbulence kinetic energy and specific dissipation rate must be non-negative quantities. The initial turbulent viscosity field computed using any eddy viscosity model will be non-negative. The optimization procedure can be greatly simplified by updating the log of the turbulent viscosity field. Updating $\log(\nu_T)$ automatically enforces the non-negativity constraint,

so the resulting optimization problem is unconstrained. This both simplifies the optimization procedure and allows us to try a larger range of optimization methods. The transformation of the sensitivity gradient is computed by simply multiplying the sensitivity gradient computed using the adjoint method by the turbulent viscosity:

$$\frac{\partial J}{\partial \log(\nu_T)} = \nu_T \frac{\partial J}{\partial \nu_T}.$$

3.3.1 L-BFGS method

Since the adjoint method provides only gradient information at a particular turbulent viscosity field, a quasi-Newton method is a good option for performing the optimization. Quasi-Newton methods construct an approximation to the Hessian matrix using only the sensitivity gradient. Using the additional information provided by the approximate Hessian matrix greatly accelerates convergence, especially once the gradient has been sufficiently reduced. The number of degrees of freedom in the turbulent viscosity field is typically large, especially for the two-dimensional case. The full approximate Hessian matrix is dense with the same number of rows and columns as the number of degrees of freedom in the problem, and the required memory for storing the approximate Hessian matrix can thus be very large. To reduce the memory requirements, the low-memory extension of the Broyden-Fletcher-Goldfarb-Shanno (L-BFGS) algorithm is used. This method computes an approximation to the Hessian matrix using only the gradient and position information at a small number of previous iterations, continuously replacing the information obtained at the oldest iteration with information from the current iteration. Furthermore, the inverse of the approximate Hessian matrix can be updated very efficiently using the Sherman-Morrison formula, since the update only involves adding a rank one matrix to the approximate Hessian [11]. The L-BFGS method thus allows us to accelerate the convergence of the optimization without dramatically increasing the computational or memory cost. For this work, the NLOpt library, which includes an efficient implementation of the L-BFGS algorithm, is used to perform the optimization [6].

3.3.2 Statistical modeling of structural uncertainties

The inverse modeling step described above computes a true turbulent viscosity field, which is denoted as ν_T^* . The goal is to construct a statistical model of the discrepancy between the true turbulent viscosity field and that predicted using the $k - \omega$ model, which is denoted as $\nu_T^{k-\omega}$. Specifically, the log-discrepancy in the turbulent viscosity field, denoted as $X = \log(\nu_T^*) - \log(\nu_T^{k-\omega})$, is modeled as a zero mean stationary Gaussian random field. The log-discrepancy is modeled to ensure that turbulent viscosity field generated by sampling X is nonnegative. The spatial correlation of this field is described using a covariance function. The squared exponential covariance function is chosen as the covariance function and is given by:

$$\text{cov}(y_i, y_j) = \sigma^2 \exp \left(\frac{-(\log(y_i) - \log(y_j))^2}{2\lambda^2} \right),$$

where y_i and y_j are spatial coordinates. The parameters σ and λ are not known a priori, but must be determined using statistical analysis. The squared exponential covariance function represents the belief that the log-discrepancy varies smoothly in space.

Maximum likelihood estimation (MLE) is used to estimate the parameters of the covariance function. This approach seeks to determine the set of parameters that is most likely to have generated the observed turbulent viscosity discrepancy. Since the discrepancy is modeled as a Gaussian random field, the probability density function of the discrepancy is described by a zero mean multivariate Gaussian, that is:

$$f_X(x|\sigma, \lambda) = \frac{1}{(2\pi)^{k/2} |\Sigma(\sigma, \lambda)|^{1/2}} \exp \left(-\frac{1}{2} x^T \Sigma(\sigma, \lambda)^{-1} x \right),$$

where $\Sigma(\sigma, \lambda)$ is the covariance matrix, and k is the dimension of the random vector of discrepancies X , i.e. the number of nodes in the mesh. The likelihood function L can be thought of as the unnormalized probability distribution of the parameter set taking particular values, conditioned on the observed data x , and is computed

directly from the conditional probability $f_X(x|\sigma, \lambda)$ [10]:

$$L(\sigma, \lambda|x) = f_X(x|\sigma, \lambda).$$

Here, x is the observed turbulent viscosity log-discrepancy field. To determine the parameter set (σ, λ) that is most likely to have generated the realized discrepancy field, the parameter set that maximizes the likelihood function is determined. For computational convenience, the log-likelihood function $\log(L)$ is maximized. Since the log-likelihood is monotonically related to the likelihood function, it is unimportant which function is maximized.

3.3.3 Propagation of structural uncertainties

Quantifying the uncertainty in quantities of interest requires propagation of the uncertainty in the turbulent viscosity field. For simplicity, non-intrusive techniques are used to perform the uncertainty propagation. This involves sampling the statistical model to produce input parameter samples and computing the quantities of interest for these samples. The model outputs computed for these sample inputs are then used to estimate the statistics, such as the mean and variance, of the quantities of interest. The advantage of non-intrusive techniques is that they do not require modification of the simulation code to compute the statistics of the outputs. Non-intrusive methods can be “wrapped around” the simulation code, providing the sample inputs and processing the simulation code outputs to estimate the statistics. This greatly simplifies the process of estimating the output statistics. In this work, the Monte Carlo method is used to compute the statistics of the mean flow field. The model of the discrepancy in the turbulent viscosity field is sampled N times, and the mean flow field is computed and stored for each sample. The expectation of the mean flow field is estimated as

$$\mathbb{E}[u(y)] = \frac{1}{N} \sum_i u_i(y).$$

The variance of the mean flow field is estimated as

$$Var(u(y)) = \frac{1}{N} \sum_i \left(u_i(y) - \frac{1}{N} \sum_i u_i(y) \right)^2.$$

To generate samples of the turbulent viscosity field, the Karhunen-Loève (K-L) expansion of the log-discrepancy Gaussian random field is computed. The K-L expansion is a spectral decomposition of a random process involves computing the spectral decomposition of the covariance kernel. The advantage of the K-L expansion is that this spectral decomposition decomposes the random field into the product of deterministic, spatially varying modes and independent, identically distributed (i.i.d.) random variables. Once the characteristic modes have been computed, one only needs to generate i.i.d. samples of a random variable, which is relatively straight forward.

For a given geometry, the discrete K-L expansion of the random field is computed as:

$$X(y, \theta) \approx \sum_{i=1}^{N_{K-L}} \sqrt{\lambda_i} x_i(y) \phi_i(\theta),$$

where the $(\lambda_i, x_i(y))$ are eigenvalue/eigenvector pairs of the covariance matrix, and $\phi_i(\theta) \sim \mathcal{N}(0, 1)$ are i.i.d. normally distributed random variables with mean zero and unit variance [8][1]. The number of K-L modes N_{K-L} used to construct the K-L expansion depends on the decay rate of the λ_i , which is controlled by the choice of covariance kernel. The smoother the covariance kernel, the more rapidly the λ_i decay. Since the log-discrepancy typically varies smoothly in space, the full K-L expansion can be approximated quite well with very small N_{K-L} .

3.4 Numerical results

The numerical results presented in this section are for flow through a periodic straight walled channel at $Re_\tau = 180$, which approximately equates to $Re = 5,600$ based on the channel height. The turbulent viscosity inversion and statistical modeling are performed by considering this flow case. The uncertainty propagation is then

performed by considering flows at higher Reynolds numbers to test the validity of the statistical model.

3.4.1 Turbulent viscosity inversion

Figure 3-4 shows the results of the optimization procedure for the straight walled channel. The objective function value decreases from an initial value $\mathbf{J} = 6.3127 \times 10^{-1}$ to $\mathbf{J} = 4.6796 \times 10^{-6}$ after 100 optimization iterations. The initial velocity profile predicted by the Wilcox $k - \omega$ model is lower everywhere except very close to the wall in the log law region, with a maximum relative error of approximately 10%. The optimized velocity profile matches the DNS velocity profile very well, with a maximum relative error of approximately 1%. The figure on the right depicts the initial and optimized turbulent viscosity profile. The DNS viscosity profile represents the effective turbulent viscosity computed using a simple force balance relation:

$$\nu_{T,\text{eff}} = \left[\frac{1}{(1 - y/\delta)} \frac{\partial U_{DNS}}{\partial y} \right]^{-1},$$

where the velocity gradient values have been provided in the DNS database. The optimized turbulent viscosity profile is nearly identical to the DNS effective turbulent viscosity, even near the channel centerline where the solution is relatively insensitive to changes in the turbulent viscosity. The path taken by the L-BFGS algorithm is plotted in figure 3-3. The objective function is steadily reduced until the optimized RANS profile matches the DNS profile.

It is important to note the importance of the regularization term for this problem. The form of the sensitivity gradient and the homogeneous Neumann boundary condition enforced at $y/\delta = 1$, which arises due to the symmetry of the problem, imply that the sensitivity gradient of $\mathbf{J}$ at the channel centerline is identically zero. Physically, this agrees with the intuition that changing the viscosity in regions where the velocity gradient is zero does not affect the resulting flow field. This means that the optimization routine will never change the value of the turbulent viscosity at $y/\delta = 1$, and the resulting optimization problem is ill-posed. This ill-posedness manifests itself

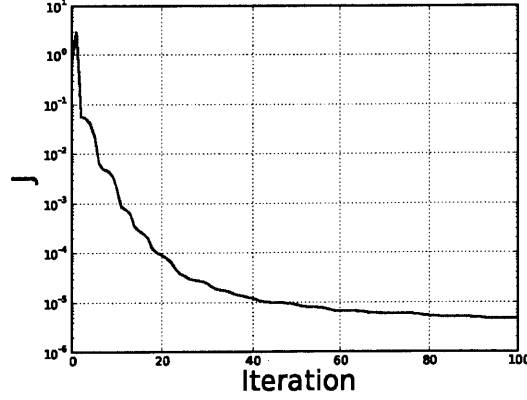
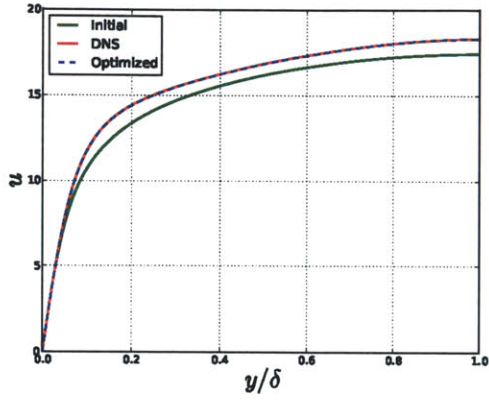
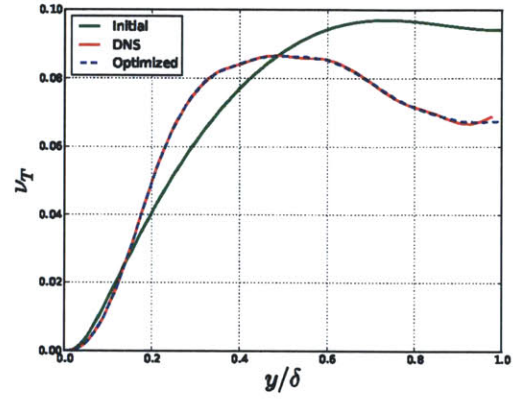


Figure 3-3: Objective function values during optimization.

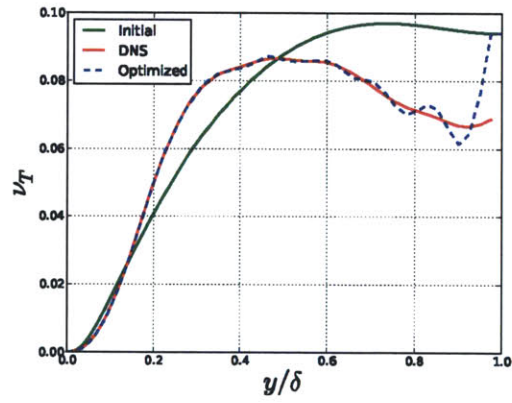
in the form of oscillations in the optimized turbulent viscosity profile near the channel centerline. The plot shown at the bottom of figure 3-4 demonstrates this issue. The optimized turbulent viscosity profile shows good agreement until $y/\delta = 0.4$, where oscillations appear and grow up to $y/\delta = 1.0$. Since the velocity gradient is small in the region $0.4 < y/\delta < 1.0$, the oscillations in the viscosity field do not significantly affect the computed velocity profile. However, since the statistical model is used to predict the discrepancy in the turbulent viscosity field, these oscillations will have a large impact on the statistical model. The regularization term remedies this issue by introducing a nonzero gradient at $y/\delta = 1$. To determine the proper value of the regularization parameter, the value of ε was increased until significant improvement was made in the agreement between the DNS effective and RANS optimized viscosity fields after 100 optimization steps. Ultimately, a value of $\varepsilon = 1.0 \times 10^{-4}$ was selected. As seen in figure 3-3, most of the change in the objective function J is made during the first fifty optimization iterations, where the magnitude of J is much larger than ε . Once the DNS and RANS velocity profiles match and J is small compared to ε , the regularization term becomes dominant, and further iterations damp the oscillations in the viscosity field.



(a) Velocity profile



(b) Viscosity profile with regularization



(c) Viscosity profile without regularization

Figure 3-4: Initial and optimized velocity and viscosity profiles compared to DNS results.

3.4.2 Statistical modeling for the straight walled channel

The results of the RANS inverse problem presented above were used to construct the statistical model using maximum likelihood estimation to estimate the parameters of the covariance function. Figure 3-5 shows the spatial variation of the log-discrepancy in the turbulent viscosity versus $\log(y/\delta)$. To model the field depicted in figure 3-5,

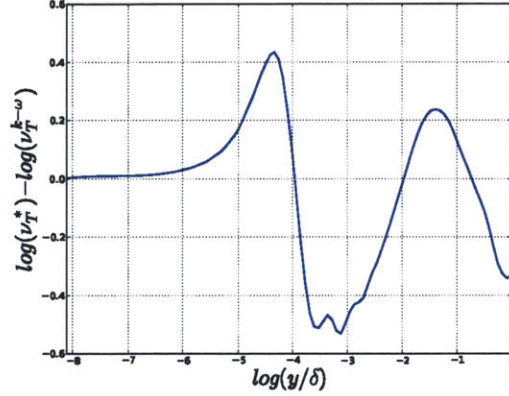


Figure 3-5: Spatial variation of the turbulent viscosity log-discrepancy.

the set of parameters that maximizes the log-likelihood function must be determined. The log-likelihood function is computed as

$$\log(L) = -\frac{1}{2} \sum_{i=1}^N \left[\log(\sigma_{\Sigma}^i) - \frac{(X^T v_{\Sigma}^i)^2}{\lambda_i} \right],$$

where σ_{Σ}^i and v_{Σ}^i are the singular values and singular vectors of the covariance matrix, respectively. Clearly, if any of the singular values of Σ are zero, the value of $\log(L)$ is not well-defined. To address this issue, it is assumed that a small error e has been made in the estimation of the true turbulent viscosity field, so that the log-discrepancy is actually given by

$$X = \log \left(\frac{\nu_T^* + e}{\nu_T^{k-\omega}} \right) \approx \log \left(\frac{\nu_T^*}{\nu_T^{k-\omega}} \right) + \frac{e}{\nu_T^*}.$$

In computing the log-likelihood function, the term $(e/\nu_T^*)^2$ is added to the diagonal of the covariance matrix Σ , since the error term relates to the variance of the Gaussian

field. The value of e is chosen to be small relative to the largest singular value of Σ . A value of $e = 10^{-6}$ was used as it satisfies this requirement. Decreasing e below this value does not change the estimated parameter set.

In general, the log-likelihood function is nonlinear in the parameter set. In that case, determining the parameter set that maximizes the log-likelihood requires some sort of gradient-free optimization method. For this work, since the dimension of the parameter set is small, the log-likelihood function is plotted for a large number of parameter sets and observe where the maximum value occurs. Figure 3-6 shows a plot of the log-likelihood function as a function of the parameter set (σ, λ) . The parameter set $(\sigma, \lambda) = (0.1898, 0.1532)$ maximizes the log-likelihood function, and this set is used in the statistical model.

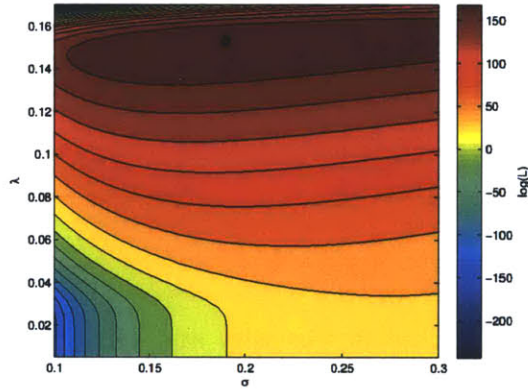


Figure 3-6: Contours of log-likelihood function, showing maximum value at $(\sigma, \lambda) = (0.1898, 0.1532)$.

3.4.3 Uncertainty propagation

For each friction Reynolds number considered, 500 Monte Carlo simulations were performed to propagate the uncertainty. Sample turbulent viscosity profiles are generated by sampling from the Gaussian random field with the parameter set determined using MLE. Figure 3-7 shows five sample turbulent viscosity profiles and the corresponding sample velocity profiles for flow at $Re_\tau = 180$. The turbulent sample viscosity fields vary smoothly in space. Figure 3-8 shows the mean and variance of the computed

samples. The solid blue line represents the mean velocity profile computed from the Monte Carlo samples. The DNS velocity profile mostly falls within the 2σ error bars (the shaded pink regions). The error bars grow larger towards the channel centerline, reflecting the fact that the level of uncertainty in the velocity profile near the wall is small relative to the uncertainty near the centerline. This agrees with the results presented in figure 3-4, which show that the velocity discrepancy between the RANS and DNS solution is small very near the wall, and remains nearly constant outside of this region.

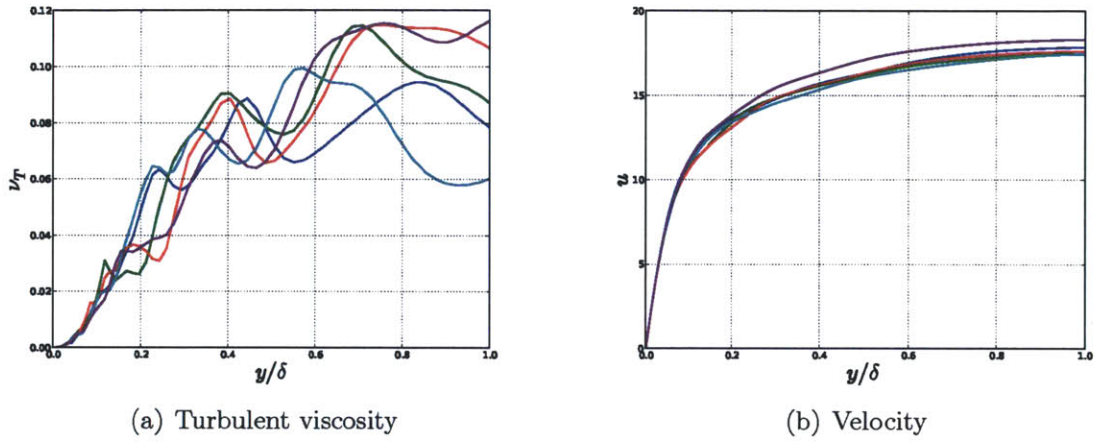
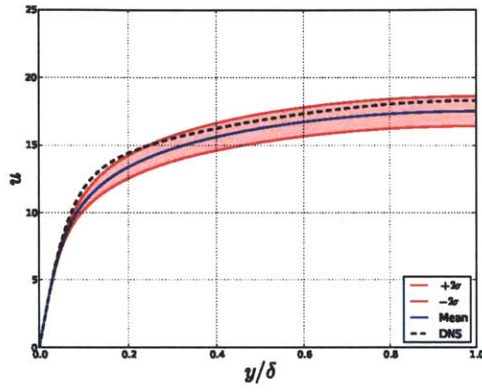
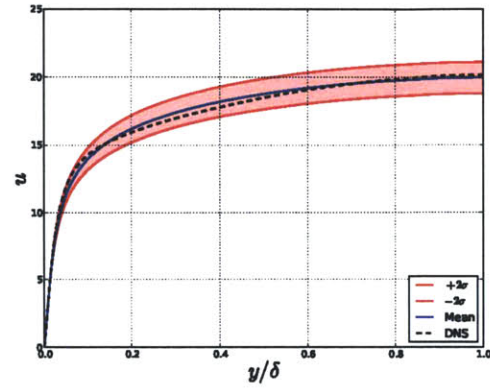


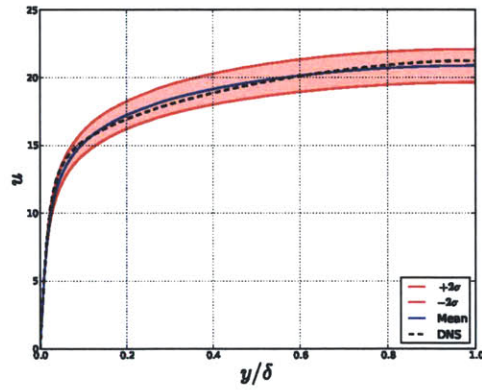
Figure 3-7: Realizations of turbulent viscosity and velocity from Monte Carlo simulation at $Re_\tau = 180$.



(a) $Re_\tau = 180$



(b) $Re_\tau = 395$



(c) $Re_\tau = 590$

Figure 3-8: Monte Carlo simulation results for three friction Reynolds numbers.

Chapter 4

Quantifying turbulence model uncertainty for 2-D flows

4.1 Introduction

In this chapter, the framework described previously is extended to more complex flows. This extension provides a statistical model of structural uncertainty that allows for uncertainty quantification of RANS simulations of general turbulent flows. To extend the method to more complex flows, the statistical model is constructed by considering flow through a collection of randomly generated 2-D geometries. The adjoint method is again used to solve an inverse problem for each of the random geometries. The inversion process is depicted in figure 4-1. For each geometry considered, the DNS and RANS flow solutions are computed on the appropriate meshes. The DNS flow field is used to compute the true RANS turbulent viscosity field using the adjoint optimization framework described previously. The optimized turbulent viscosity field and flow solution are then used to construct a statistical model of the structural uncertainties which can in turn be used to propagate uncertainty to the quantities of interest.

Once the inverse problem has been solved for each random geometry, the statistical model of uncertainty is constructed. The discrepancy between the RANS turbulent viscosity field and the true turbulent viscosity field is represented as a Gaussian ran-

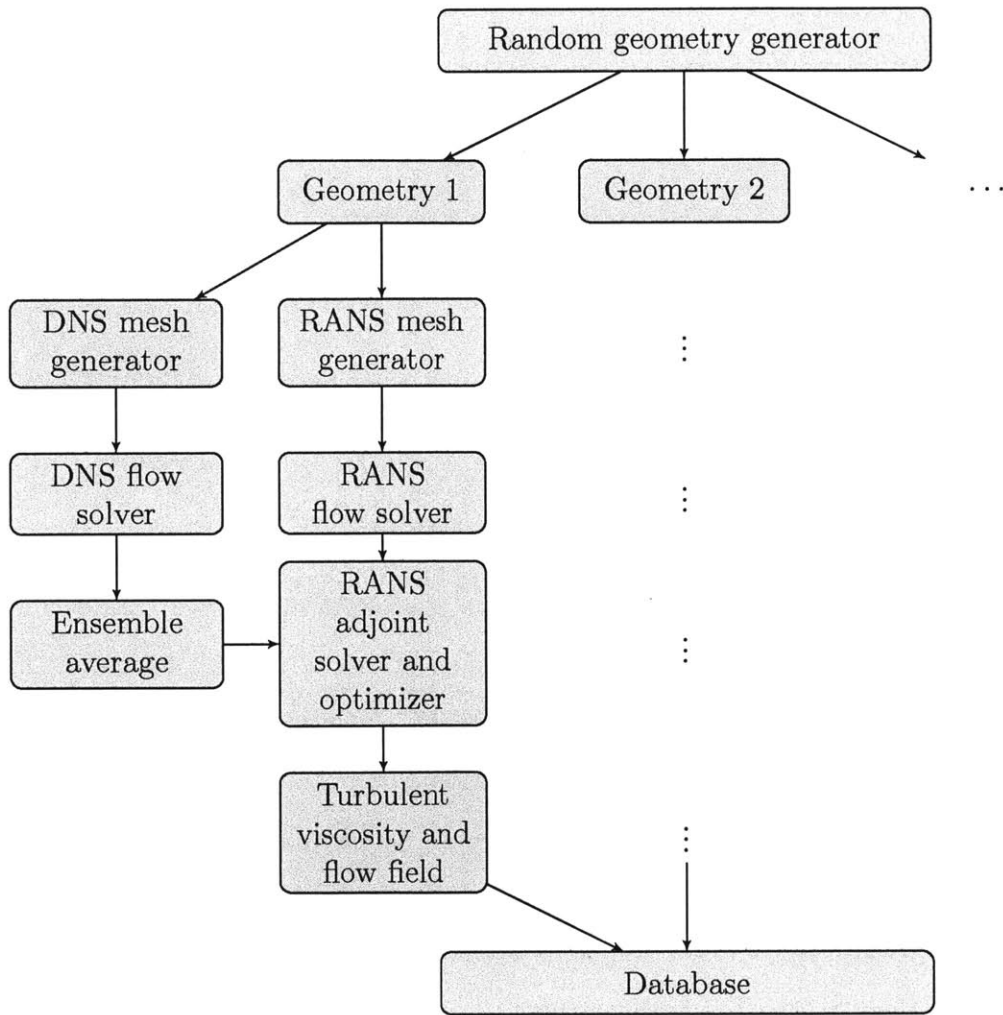


Figure 4-1: Flow chart describing the turbulent viscosity field inversion

dom field. Given a RANS flow solution, this model is sampled to produce realizations of the turbulent viscosity field with spatial distributions of discrepancy from the computed RANS turbulent viscosity field that are statistically similar to those observed for the DNS flow solutions. Flow solutions are computed for each turbulent viscosity realization, and are then used to estimate the uncertainty in the quantities of interest.

4.2 Random geometry generation

The geometries used to construct the database of flows must satisfy two important conditions:

1. They must be sufficiently simple. The direct numerical simulation requires an extremely fine mesh to resolve the relevant scales of turbulent motion. Using simple geometries reduces the complexity of the resulting meshes.
2. They must produce flow phenomena observed in complex engineering applications. Since the flows stored in this database are used to construct a statistical model for structural uncertainties arising in complex flows, they should exhibit similar flow characteristics, including regions of separation, recirculation, and reattachment.

To satisfy these conflicting requirements, a random channel geometry generator was developed. The channel walls are generated by simulating a Gaussian process with a correlation function

$$C(d) = \exp(-d^2/(c_0^2 + c_1|d|)).$$

This correlation function was chosen as it produces smoothly varying wall geometries. The Gaussian process is conditioned to have zero slope at the inlet and outlet sections of the channel, and is simulated using the matrix factorization method [3].

Unstructured meshes are used to compute the RANS and DNS solutions. Near the solid boundaries, the mesh is refined to resolve the boundary layer. The interior of the domain is discretized with triangles. Two example meshes used for computing the

DNS solution are shown in figure 4-2. The meshes used to compute the RANS solutions are roughly twice as coarse as those depicted in figure 4-2. The solid boundaries are the upper and lower curved surfaces. Since the flow is computed on a periodic domain, the inlet and outlet mesh faces are identical. Since turbulence is inherently three-dimensional in nature, the meshes used to perform the direct numerical simulations must be three-dimensional. The two-dimensional meshes are translated in the z -direction to create a three-dimensional mesh.

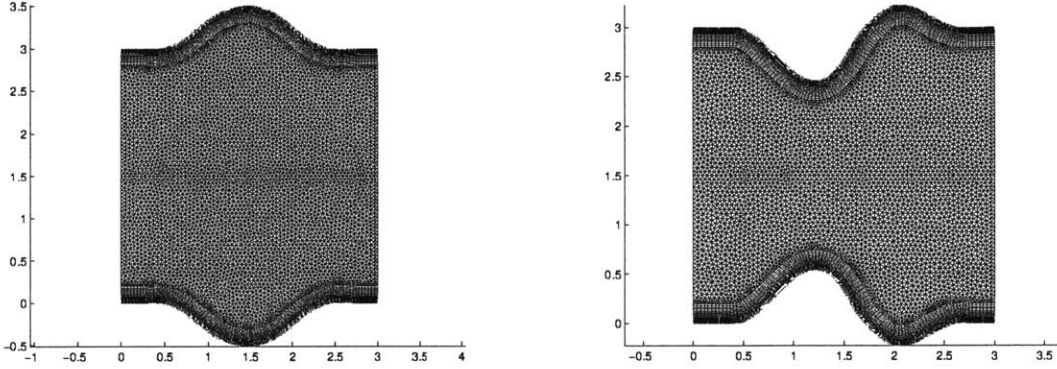


Figure 4-2: Sample DNS meshes

4.3 RANS and DNS solvers

To compute the RANS mean flow field and turbulent viscosity field, the “Joe” flow solver from Stanford’s Center for Turbulence Research was used. This code solves the compressible RANS equations on unstructured meshes using a second order accurate finite volume scheme, and includes a number of RANS turbulence models. All RANS solutions were computed using the Wilcox $k - \omega$ two-equation model, one of the most popular RANS turbulence models used in industry [17]. A unit body force in the positive x direction is applied to drive the flow, and the laminar viscosity was set to $\nu_\ell = 2.0 \times 10^{-3}$.

The CDP code, also developed at the CTR, was used to perform the direct numerical simulations. This code uses a second order accurate node based finite volume method, and handles unstructured meshes. The flow solution is advanced in time

using the Crank-Nicolson scheme, and the code is fully parallel. The spatial discretization scheme is based upon simplex superposition, which reduces the numerical dissipation introduced in the discretization [4]. Unlike the Joe code, CDP is an incompressible code. Flows with very low Mach number (on the order of $M = 0.1$) are considered to ensure that the effects of compressibility are minimal, allowing the comparison of the velocity fields computed by Joe and CDP. To compare the results of DNS to the mean velocity field computed using RANS, the unsteady DNS velocity field must be averaged. The averaging procedure is performed both in space and in time. As described above, fully three-dimensional DNS meshes are created by extruding the two-dimensional mesh in the z -direction. A spatial averaging is performed by averaging the flow field over all of the two-dimensional “slices” created by translating the mesh. Since the geometries are symmetric about the centerline, the flow solution is again averaged by averaging the values above and below the centerline. The spatially averaged solution is averaged in time over 50,000 iterations with a fixed timestep of $\Delta t = 7.5 \times 10^{-4}$ to produce the mean velocity field u_{DNS} . The same laminar viscosity and forcing used to compute the RANS solution was used for the DNS simulations.

4.4 The adjoint solver

The adjoint solver used in this work solves the continuous adjoint equations derived in chapter 2. The numerical scheme for solving the adjoint system is nearly identical to the numerical scheme used by CDP. Specifically, the scheme is second order in space and time, and uses the Crank Nicolson for time integration. To verify that the adjoint solver is computing the sensitivity gradient accurately, a verification study was performed. The adjoint solver was verified by comparing the sensitivity gradient to results computed using finite differences and a tangent flow solver. The tangent flow solver can be used to compute the velocity and pressure perturbation fields that result from introducing a perturbation in the turbulent viscosity field by solving the

linearized mean flow equations

$$\frac{\partial \delta u}{\partial t} + \mathcal{L}_{(u, \nu_T)} + \nabla \delta p = 0 \quad (4.1)$$

$$\nabla \cdot \delta u = 0$$

The numerical scheme for solving the linearized mean flow equations is similar to the scheme used to solve the continuous adjoint equations.

To compare the tangent and adjoint solvers to the finite difference method, the change in the objective function

$$\mathbf{J}(u(\nu_T)) = \|u(\nu_T) - u_{DNS}\|_{L^2}^2$$

is computed for a prescribed perturbation in turbulent viscosity field. The change in the objective function computed using the finite difference method can be computed as

$$\delta \mathbf{J}_{FD} = \mathbf{J}(u(\nu_T + \delta \nu_T)) - \mathbf{J}(u(\nu_T))$$

where $\delta \nu_T$ is a small perturbation in turbulent viscosity field. Similarly, the change in the objective function computed by solving the linearized mean flow equations is

$$\delta \mathbf{J}_{Tan} = \mathbf{J}(u(\nu_T) + \delta u) - \mathbf{J}(u(\nu_T))$$

where δu is the velocity perturbation field computed by solving equation 4.1 with a specified perturbation in the turbulent viscosity field. To verify that the adjoint sensitivity gradient is being computed correctly, $\delta \mathbf{J}_{FD}$ and $\delta \mathbf{J}_{Tan}$ are compared to the change in the objective function computed by integrating the adjoint sensitivity gradient over the domain:

$$\delta \mathbf{J}_{Adj} = \iiint_{\Omega} \frac{\partial \mathbf{J}}{\partial \nu_T} \delta \nu_T \, dx$$

As an example case, a Gaussian bump perturbation is prescribed in the turbulent

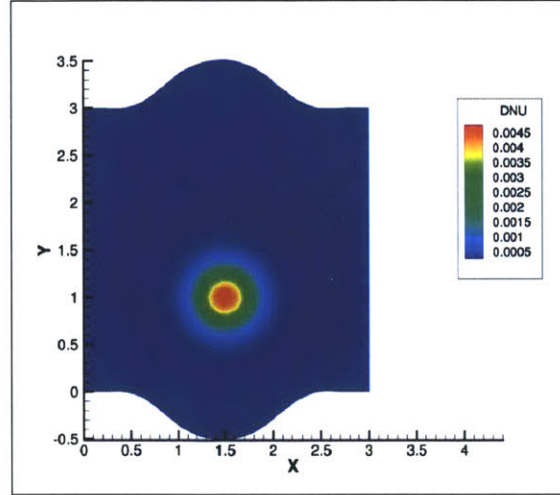


Figure 4-3: Turbulent viscosity perturbation field

viscosity field:

$$\delta\nu_T(x, y) = 0.005e^{-[(x-1.5)/0.3]^2 - [(y-1.0)/0.3]^2}.$$

The magnitude of the perturbation is chosen to be small relative to the magnitude of the underlying turbulent viscosity field. The perturbation field is plotted in figure 4-3. The solution of the linearized mean flow equations is plotted in figures 4-5 and 4-6. The flow solution about which the linearization is performed is shown in figure 4-4. The left shows the result computed using finite differences, i.e. $u(\nu_T + \delta\nu_T) - u(\nu_T)$, and the right shows the perturbation field computed by solving equation 4.1. There is excellent agreement between the two perturbation fields computed using finite differences and by solving the linearized mean flow equation. The blue regions show where the mean velocity decreases, and the red regions indicate where the mean velocity increases. In the region where the perturbation is largest, the velocity gradient is small. The perturbation in the velocity field is thus caused principally by the gradient in the turbulent viscosity perturbation field. Where $\partial\delta\nu_T/\partial y$ is negative, the axial velocity is expected to decrease, as is observed in figure 4-5.

Table 4.1 shows the numerical values of the change in the objective function computed using the three methods described above, as well as the percent error between the finite difference value and the tangent and adjoint values. The agreement between the three values is excellent, indicating that the adjoint sensitivity gradient is

	δJ	% Error
Finite Difference	-7.661×10^{-3}	—
Tangent	-7.545×10^{-3}	1.51
Adjoint	-7.650×10^{-3}	0.14

Table 4.1: Comparison of objective function change

being computed correctly. Changing the turbulent viscosity perturbation to different distributions results in similar agreement between the change in the objective function values.

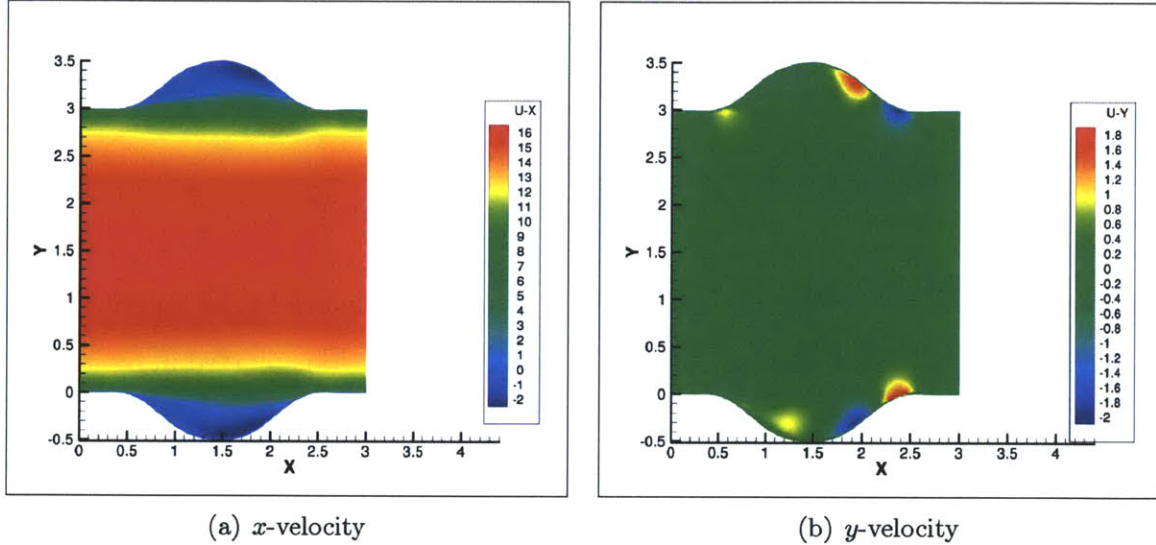
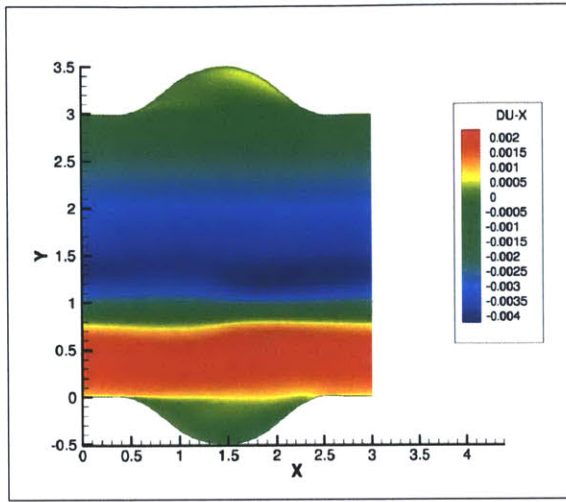


Figure 4-4: Baseline velocity fields

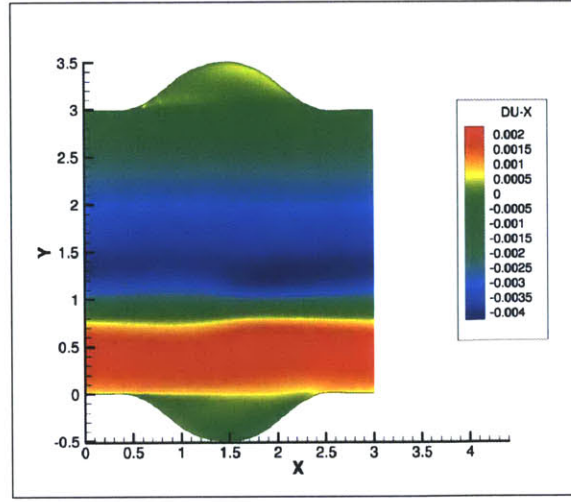
4.5 Numerical results

4.5.1 Comparison of RANS and DNS results

For each of the geometries considered, the RANS equations were solved, and the turbulent viscosity profile computed using the $k - \omega$ model was stored for use in the optimization step. The mean DNS flow field was also computed and stored. In all simulations, a unit body force is applied in the positive x direction to drive the flow. Figure 4-7 shows a comparison between the mean velocity fields computed using direct numerical simulation and by solving the RANS equations for one of the

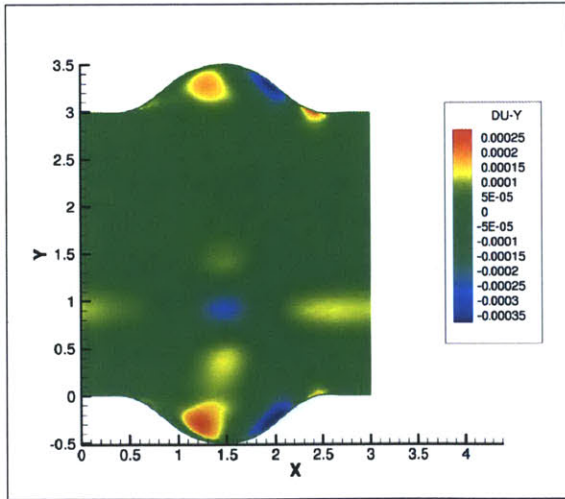


(a) Finite Difference

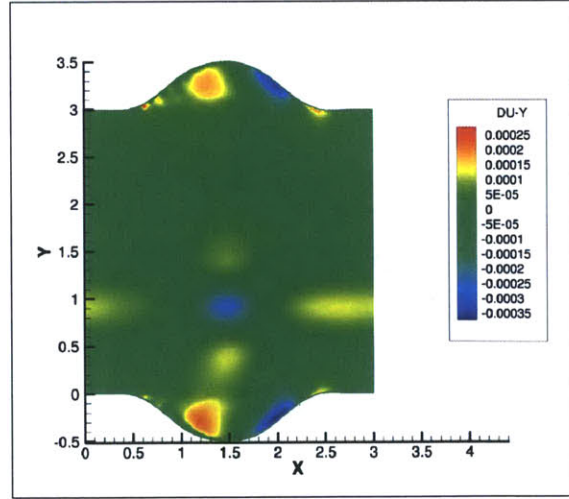


(b) Tangent

Figure 4-5: x -velocity perturbation field



(a) Finite Difference



(b) Tangent

Figure 4-6: y -velocity perturbation field

geometries considered. The velocity fields computed using the two methods exhibit the same basic flow features, including the complex region of recirculation that forms as a result of the “bump” in the center of the geometry. However, while the y -velocity fields computed using the two methods agree fairly well, the x -velocity component shows a large discrepancy, especially near the center of the channel. The largest discrepancy in the x -velocity is approximately 30% of the RANS velocity. Since the RANS velocity is larger than the mean DNS velocity in most of the channel, it is apparent that the $k - \omega$ model is underestimating the level of turbulent dissipation in this flow.

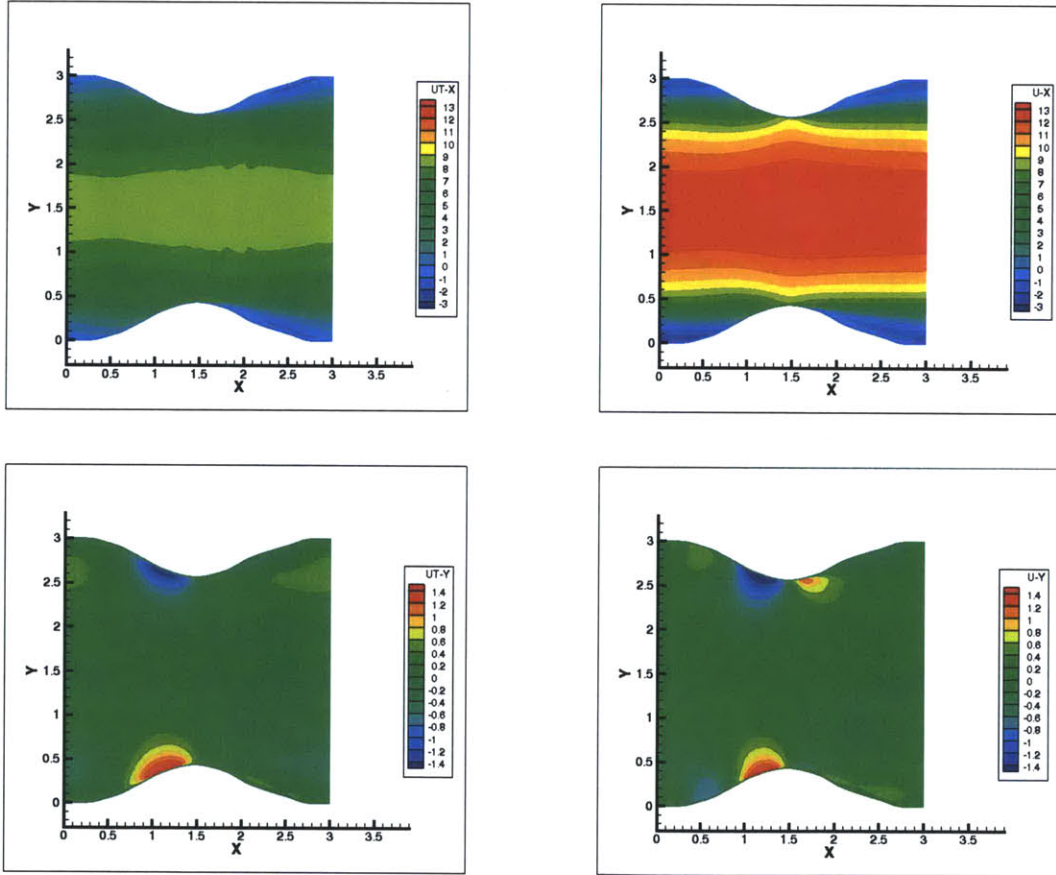


Figure 4-7: Comparison between the mean DNS (left) and RANS (right) x -velocity fields (upper) and y -velocity fields (lower).

4.5.2 Results for the RANS inverse problem

The initial turbulent viscosity field computed by solving the RANS equations was optimized to determine the turbulent viscosity field ν_T^* that produces a mean velocity field that agrees more closely with the mean DNS velocity field. Figure 4-8 depicts the mean velocity field produced by prescribing the optimized turbulent viscosity field for the same geometry depicted in figure 4-7. Comparing figures 4-7 and 4-8, it is clear that the optimized mean velocity field computed using ν_T^* shows much better agreement with the mean DNS velocity field than the original velocity field computed using the $k-\omega$ model. The regions where the flow is reversed (blue regions in the x -velocity contours) still show some disagreement after the turbulent viscosity has been optimized. Typically, the optimization of the turbulent viscosity required roughly 30 iterations to achieve a high level of agreement between the RANS and DNS flow fields, which involves tuning thousands of nodal values for the turbulent viscosity. This demonstrates the efficiency of the adjoint approach for solving large-scale inverse problems.

The level of agreement for the two velocity fields can be quantified by comparing the value of the objective function $\mathbf{J}$, which measures the difference between the RANS mean flow field and the DNS mean flow field. For the geometry shown in figures 4-7 and 4-8, the initial objective function value was $\mathbf{J}(\nu_T^{k-\omega}) = 7.35$. For the optimized turbulent viscosity field, the objective function value was $\mathbf{J}(\nu_T^*) = 0.0840$. This level of reduction was typical of the geometries considered, as indicated by table 4.2. The last column of table 4.2 indicates the percentage change in the norm of the velocity discrepancy, i.e. $1 - \sqrt{\mathbf{J}(\nu_T^*)/\mathbf{J}(\nu_T^{k-\omega})}$, which represents the percentage of the velocity discrepancy that can be attributed to uncertainty in the turbulent viscosity field.

There are a number of possible sources for disagreement between the RANS and mean DNS flow solutions. These sources include the statistical noise introduced by the averaging of the DNS solution; the effect of compressibility not captured by the incompressible DNS simulation; the differences between the numerical schemes used to compute the RANS and DNS solutions; the assumption of mean rate-of-

	$\mathbf{J}(\nu_T^{k-\omega})$	$\mathbf{J}(\nu_T^*)$	% discrepancy due to ν_T
Geometry 1	23.5	0.148	92.1%
Geometry 2	0.515	0.0449	70.5%
Geometry 3	16.2	0.254	87.5%
Geometry 4	6.19	0.0117	86.3%
Geometry 5	7.15	0.0646	90.5%
Geometry 6	0.165	0.0127	72.3%
Geometry 7	15.9	0.308	86.1%
Geometry 8	7.35	0.840	89.3%

Table 4.2: Comparison between the velocity discrepancies for the velocities computed using the $k - \omega$ model and the optimized turbulent viscosity.

strain/Reynolds stress anisotropy alignment made by the Boussinesq hypothesis; and the uncertainty in the turbulent viscosity field. The substantial reductions in the velocity discrepancy presented in table 4.2, which were obtained by only varying the turbulent viscosity field, suggest that the uncertainty in the flow solution can be largely attributed to the inability of the $k - \omega$ model to estimate the true turbulent viscosity. This supports the assumption made earlier that the discrepancy between the RANS and DNS results is primarily due to the uncertainty in the turbulent viscosity. These results also quantify the level of uncertainty introduced by the sources of uncertainty not related to uncertainty in the turbulent viscosity. Since the RANS velocity field cannot be made to match the DNS mean velocity field exactly by changing the turbulent viscosity, the other sources of uncertainty are not negligible. This level of uncertainty can be quantified by considering the discrepancy in the RANS and DNS velocity fields after the turbulent viscosity has been optimized.

For reference, the turbulent viscosity field computed using the $k - \omega$ model and the optimized turbulent viscosity profile are plotted in figure 4-9. To highlight the differences between the two fields, the log-discrepancy between the two fields, defined as $\log(\nu_T^*/\nu_T^{k-\omega})$, is plotted in figure 4-10. The log-discrepancy field depicted in figure 4-10 is typical of the geometries considered. The largest changes in the turbulent viscosity field, corresponding to the areas where the log-discrepancy magnitude is largest, are made around the “bump” in the geometry, where the flow separates from

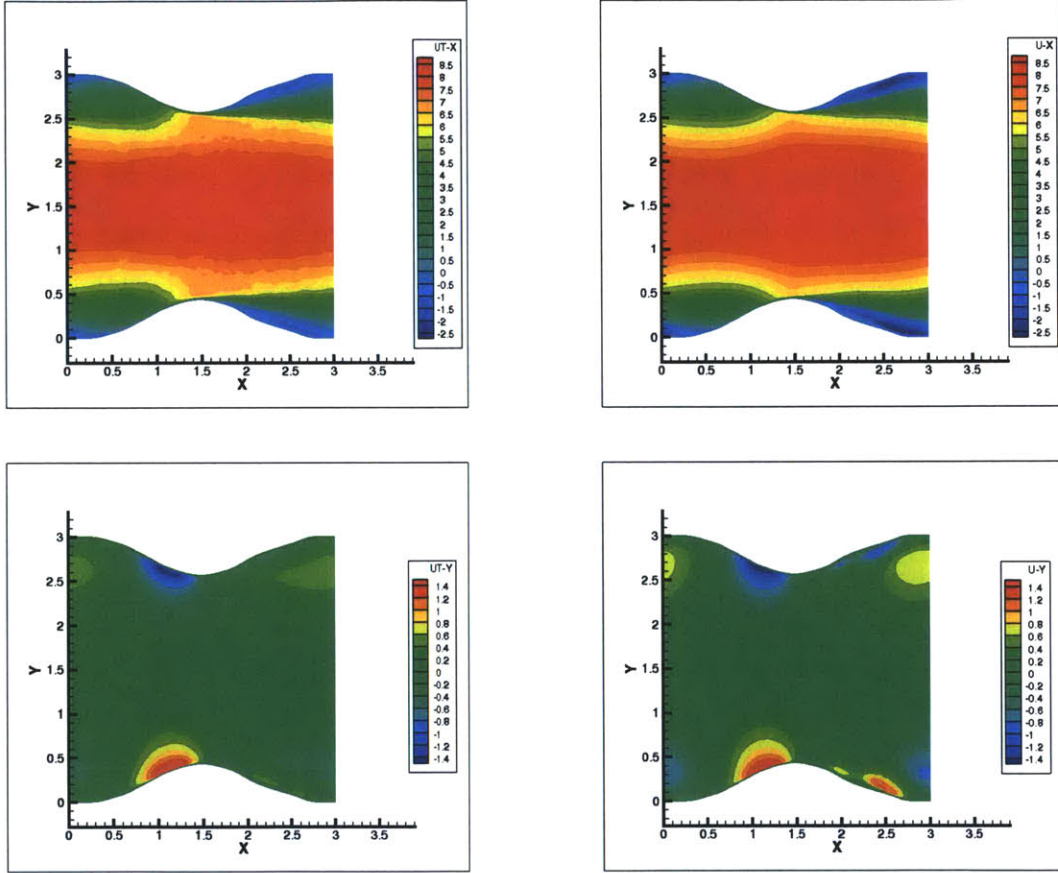


Figure 4-8: Comparison between the mean DNS (left) and optimized (right) x -velocity fields (upper) and y -velocity fields (lower).

the wall. It is clear that the presence of separation in the flow introduces a great deal of uncertainty in the estimate of the turbulent viscosity field. This field is also highly anisotropic and non-stationary. Near the wall, the correlation length between the values of log-discrepancy in the streamwise direction is much larger than in the direction normal to the solid boundary. This non-stationarity is consistent with the results for flow through a straight channel presented in the previous chapter. For the straight channel, the log-discrepancy field was highly non-stationary. Values near the wall were more highly correlated than values far from the wall, as observed here. For the geometries considered in this work, the magnitude of the variations is also much larger near the wall than it is far away from the wall. Conversely, near the centerline of the channel, the shear strain-rate is very small relative to the shear strain rate near the solid boundaries, and the corresponding log-discrepancy magnitude is small. It is clear that the flow is most sensitive to changes in the turbulent viscosity in regions where the shear strain rate is largest. These characteristics should be captured by the statistical model of the log-discrepancy.

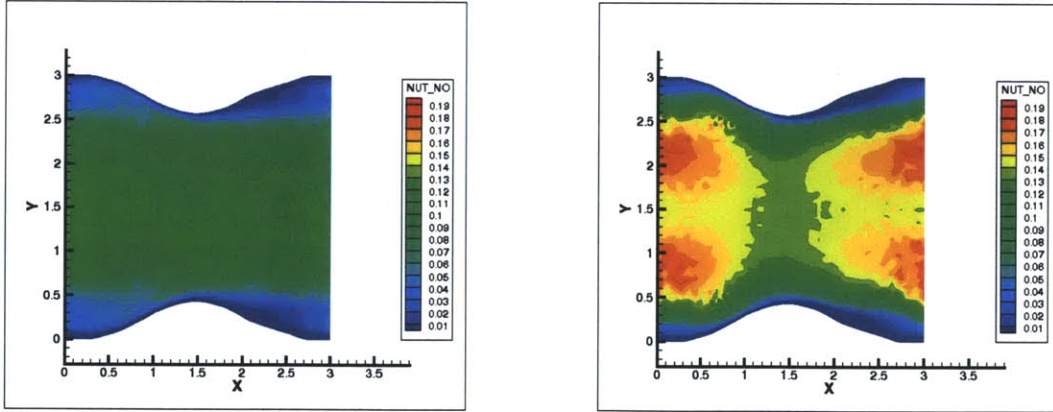


Figure 4-9: Comparison between $k-\omega$ (left) and optimized (right) turbulent viscosity fields.

4.5.3 Statistical modeling

The apparent correlation between the magnitude of the log-discrepancy in the turbulent viscosity and the shear-strain rate was captured in the statistical model by

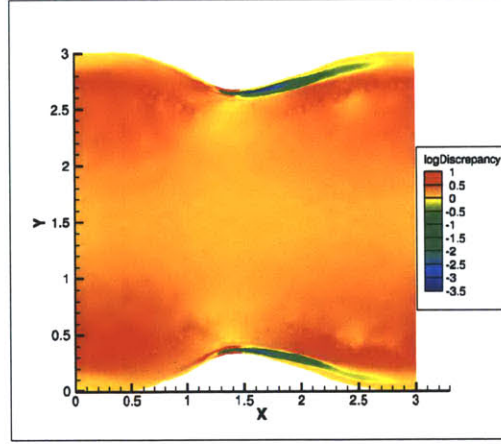


Figure 4-10: log-discrepancy between the optimized and $k - \omega$ turbulent viscosities.

performing a linear regression on the data obtained for the random geometries. Specifically, the relation between the magnitude of the log-discrepancy and the corrected velocity strain-rate norm, defined as $\|S\|_2(1 - \exp(-d/d_0))$ is used to perform the regression. Here S is the velocity strain-rate tensor, d is the distance from the solid wall, and d_0 is the distance from the wall of the point where $\|S\|_2$ is largest. The norm of the velocity strain rate measures the shear stress at a given location. A plot of the log-discrepancy versus the corrected velocity strain-rate norm is shown in figure 4-11. The magnitude of the log-discrepancy does show a positive correlation with the corrected shear strain-rate norm. The correction factor $(1 - \exp(-d/d_0))$ weights the points furthest from the wall, where $\|S\|_2$ is smallest, more heavily than points near the wall, where d is small. For points very near the wall, the magnitude log-discrepancy is very small, since the log-discrepancy must go to zero at the wall. The correction factor thus limits the impact of points with large $\|S\|_2$ and small discrepancy on the linear regression. Since the log-discrepancy must go to zero at the wall, the linear regression is constrained to pass through the origin. The resulting regression is plotted in red in figure 4-11.

The results presented in the previous section indicate that the log-discrepancy field is highly anisotropic and non-stationary, and it is insufficient to simply model this random field as stationary and isotropic. Instead, the log-discrepancy is then

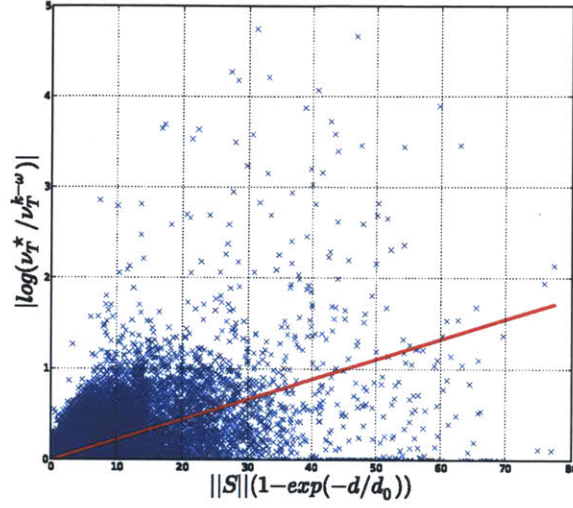


Figure 4-11: log-discrepancy plotted against the corrected velocity strain-rate norm.

model by scaling an isotropic, stationary Gaussian random field based on the local corrected velocity strain-rate norm. To generate sample log-discrepancy fields, a zero mean Gaussian random field with covariance function

$$C(\mathbf{x}_1, \mathbf{x}_2) = \exp\left(\frac{-\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2}{2\lambda^2}\right)$$

was simulated on a uniform grid with a correlation length of $\lambda = 0.2$. The field was simulated using the Karhunen-Loève expansion of the covariance matrix. Each random field realization was then interpolated to the mesh points of the channel grid. The value of the random field was scaled according to the corrected RANS velocity strain-rate norm using the linear regression estimate depicted in 4-11. Since the geometries considered were symmetric, it is expected the realizations of turbulent viscosity field to be symmetric about the center of the channel. This symmetry was explicitly enforced for all random turbulent viscosity realizations by setting the values of the log-discrepancy to be equal above and below the center of the channel. A collection of random turbulent viscosity log-discrepancies is shown in figure 4-12. For the samples shown, the locations where the magnitude of the log-discrepancy is largest correspond to locations of large RANS velocity strain-rate, i.e. around the

“bump” in the geometry at $x = 1.5$, and the log-discrepancy goes to zero along the centerline of the channel at $y = 1.5$. Also, the correlation length in the streamwise direction is much smaller than in the wall normal direction. All of these features match the observations of the log-discrepancy realizations made when comparing the $k - \omega$ and optimized turbulent viscosity fields.

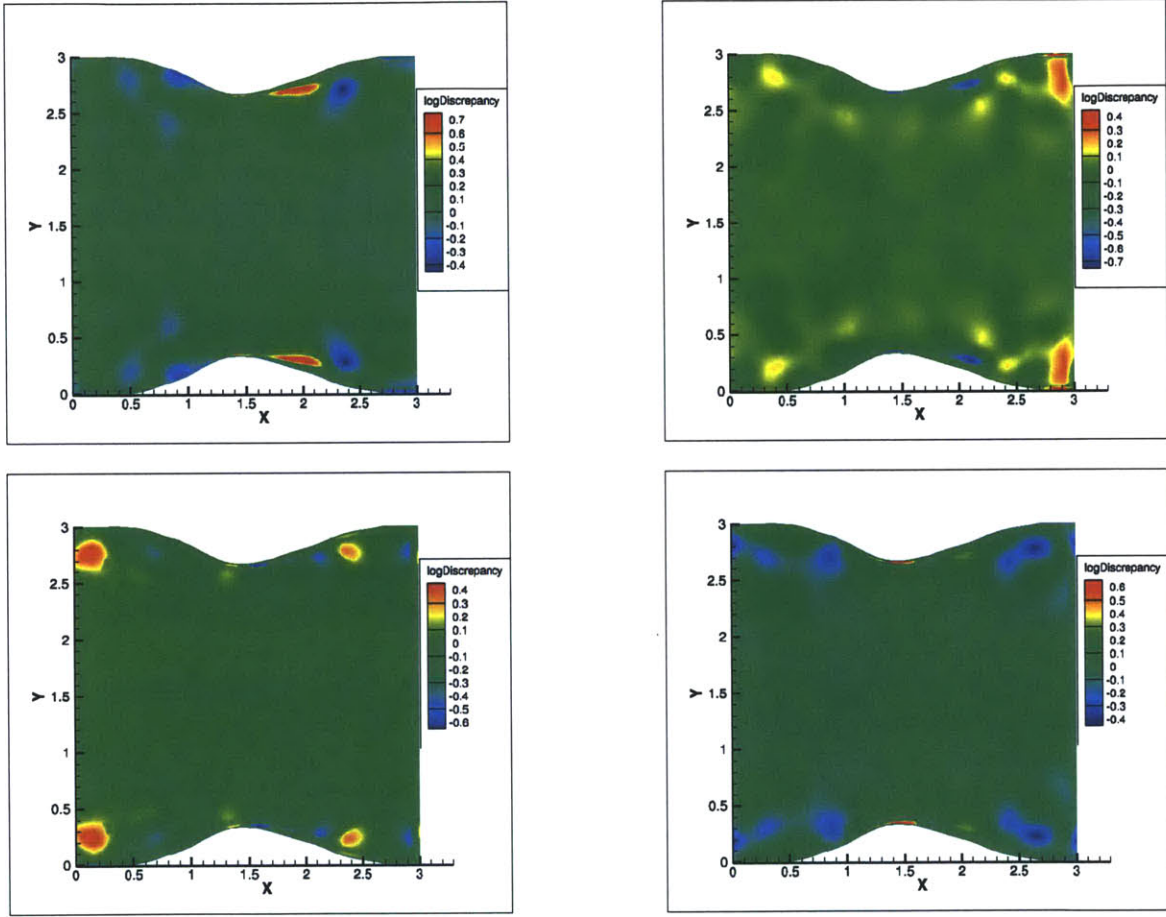


Figure 4-12: Sample realizations of the turbulent viscosity log-discrepancy field.

4.5.4 Uncertainty propagation

To propagate the uncertainty in the turbulent viscosity to the RANS velocity field, 500 Monte Carlo simulations were performed. For each Monte Carlo sample, a random log-discrepancy field was simulated using the method described above, and the corresponding sample turbulent viscosity field was computed by scaling the $k - \omega$

turbulent viscosity field. The RANS flow field was computed using the sample turbulent viscosity and stored. The mean of the Monte Carlo sample velocity fields was found to match the $k - \omega$ velocity very closely. The standard deviation of the Monte Carlo sample flow fields are plotted in figure 4-13. The region of largest variation is observed just behind the “bump” in the mesh. The large variability in the flow field is due to the relatively large uncertainty in the location of the separation point. The large variability in the turbulent viscosity discrepancy around the separation point results in uncertainty in the flow field in this region. This is consistent with the fact that RANS models typically fail to accurately estimate the separation point.

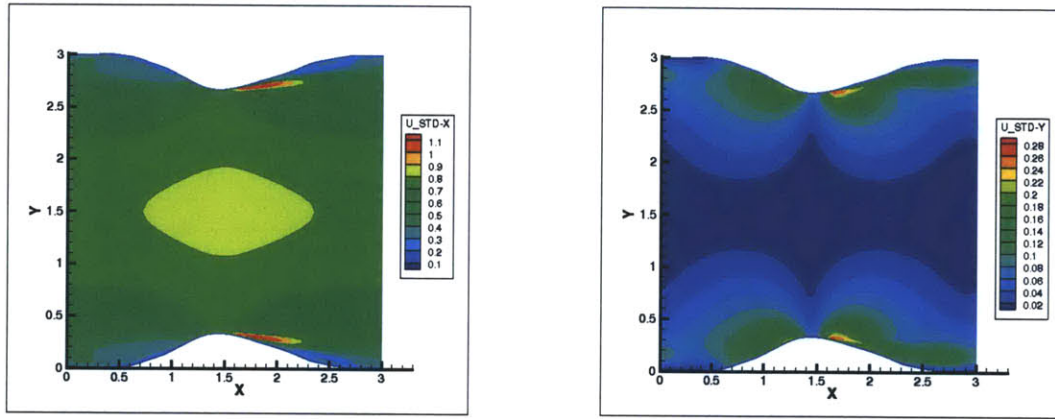


Figure 4-13: Standard deviation of the x -velocity (left) and y -velocity fields (right).

To more clearly display the Monte Carlo results, figure 4-14 shows the RANS, DNS and Monte Carlo velocity profiles at four different x locations, representing a vertical slice through the domain. The Monte Carlo profile shows the mean of the Monte Carlo sample velocity profiles, as well as the two standard deviation error intervals around the mean velocity profile, representing the 95% confidence intervals. The mean DNS x -velocity profiles typically fall outside the 2σ intervals, especially near the center of the channel. This means the estimated level of uncertainty in the turbulent viscosity is too low. However, the mean DNS y -velocity profiles are mostly contained inside the 2σ intervals. The 2σ intervals are typically largest where the $k - \omega$ and mean DNS profiles show the largest disagreement, showing that the general trend in the uncertainty is being captured. The maximum magnitude of the

random log-discrepancy samples shown in figure 4-12 are smaller than the observed log-discrepancy field shown in figure 4-10. Clearly, the model of the discrepancy fails to produce realizations with the proper discrepancy magnitude. Considering the linear fit shown in figure 4-11, the best linear fit for the relation between the discrepancy and the corrected strain-rate norm is quite flat, implying that realizations with a large discrepancy magnitude are relatively unlikely. This explains why the maximum magnitude of the log-discrepancy realizations is too low, thereby underestimating the level of uncertainty in the flow field.

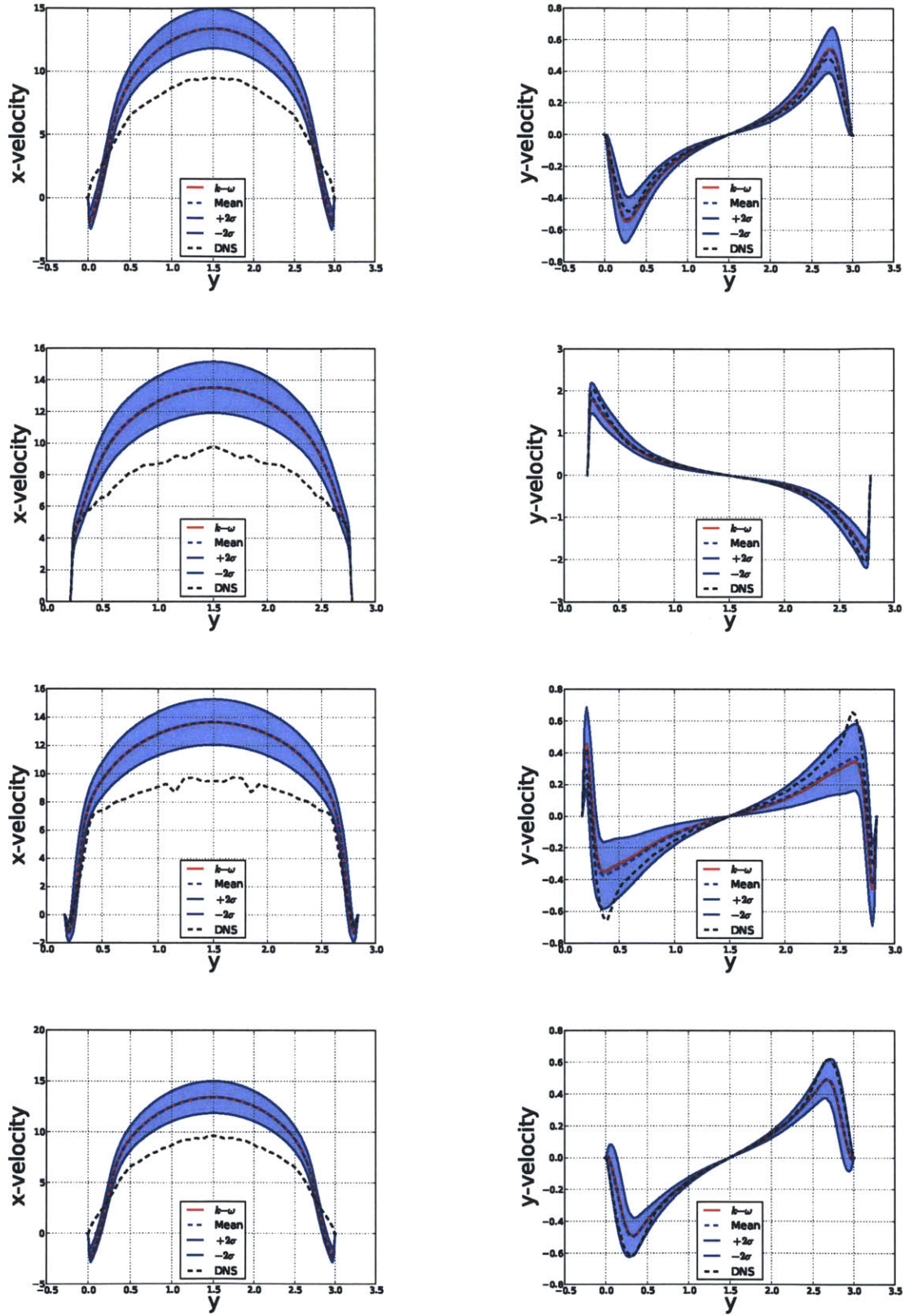


Figure 4-14: RANS, DNS, and Monte Carlo velocity profiles plotted at $x = 0.1$, $x = 1.1$, $x = 2.1$, and $x = 2.9$ (from top to bottom).

Chapter 5

Conclusions

In this thesis, a new approach for quantifying the structural uncertainties in RANS simulations has been presented. The uncertainty in the RANS flow field is attributed to uncertainty in the turbulent viscosity field estimated by the turbulence model. Numerical evidence has been provided that suggests that a significant fraction of the uncertainty in the RANS flow field can indeed be attributed to uncertainty in the turbulent viscosity field. By developing a statistical model of the uncertainty in the turbulent viscosity, the uncertainty in the quantities of interest that arises due to structural uncertainty can be estimated.

The results presented in chapter 3 clearly demonstrate the effectiveness of this framework. The level of uncertainty in the mean flow field predicted by the statistical model agrees well with the observed discrepancy between the RANS and DNS flow fields, as the DNS results are mostly contained within the 95% confidence intervals. For the 2-D simulations presented in chapter 4, the level of uncertainty predicted by the statistical model is clearly too low. The current statistical model is likely too simple to accurately capture the true statistical nature of the uncertainty in the turbulent viscosity field.

Although the results presented consider structural uncertainty in the $k - \omega$ turbulence model, the approach described in this work is entirely generalizable to any eddy viscosity model, both linear and nonlinear. The approach only requires the turbulent viscosity field computed by the turbulence model. How the turbulent viscosity field

is computed is irrelevant. Furthermore, this approach is not limited to estimating uncertainty in RANS simulations. Simulations of a variety of other physical problems include model structure uncertainty, e.g. combustion modeling and geophysical simulation. The inverse modeling procedure presented in this thesis can be applied to these problems to estimate the uncertainty in the model parameters provided an adjoint sensitivity gradient can be constructed.

There are a number of possible extensions of the work presented in this thesis. First and foremost, this framework needs to be extended to 3-D. It would also be valuable to study transonic and supersonic flows, as these regimes are of primary importance for aerospace applications. Additionally, since the relation between the Reynolds stress and the mean rate of strain is in general nonlinear, there does not always exist a turbulent viscosity field that can be prescribed to exactly predict the turbulent flow field. This motivates modeling the uncertainty in the Reynolds stress tensor rather than the turbulent viscosity field. This requires developing a statistical model for a tensor field rather than a scalar field. Also, the inverse modeling approach presented here could potentially be used to improve the performance of current turbulence models. The inverse approach allows modelers to determine a target turbulent viscosity profile that minimizes the error for a given flow field. This information could be used to correlate model parameters with the dominant flow features such that the computed turbulent viscosity field more accurately reproduces the true turbulent viscosity field.

Bibliography

- [1] H. Chen, Q. Wang, R. Hu, and P. Constantine. Conditional sampling and experiment design for quantifying manufacturing error of transonic airfoil. AIAA Paper 2011-658, 2011.
- [2] S. H. Cheung, T. A. Oliver, E. E. Pruenzio, S. Prudhomme, and R. D. Moser. Bayesian uncertainty analysis with applications to turbulence modeling. *Reliability Engineering and System Safety*, In press.
- [3] M. W. Davis. Production of conditional simulations via the LU triangular decomposition of the covariance matrix. *Mathematical Geology*, 19(2):91–98, 1987.
- [4] F. Ham. Improved scalar transport for unstructured finite volume methods using simplex superposition. In *Annual Research Briefs, Center for Turbulence Research, NASA-AMES*, pages 347–358, 2008.
- [5] A. Jameson. Aerodynamic shape optimization using the adjoint method. In *VKI Lecture Series on Aerodynamic Drag Prediction and Reduction*, 2003.
- [6] S. G. Johnson. The nlopt nonlinear-optimization package. <http://ab-initio.mit.edu/nlopt>, 2011.
- [7] W.S. Kim, S. He, and J.D. Jackson. Assessment by comparison with DNS data of turbulence models used in simulations of mixed convection. *International Journal of Heat and Mass Transfer*, 51(5):1293–1312, 2008.
- [8] O.P. Le Maître and O.M. Knio. *Spectral Methods for Uncertainty Quantification: With Applications to Computational Fluid Dynamics*, pages 17–21. Springer, New York, first edition, 2010.
- [9] R. D. Moser, J. Kim, and N. N. Mansour. Direct numerical simulation of turbulent channel flow up to $Re_\tau = 590$. *Physics of Fluids*, 11(4):943–945, 1999.
- [10] J. Myung. Tutorial on maximum likelihood estimation. *Journal of Mathematical Psychology*, 47:90–100, 2003.
- [11] J. Nocedal. Updating quasi-Newton matrices with limited storage. *Mathematics of Computation*, 35:773–782, 1980.

- [12] P. D. A. Platteeuw, Loeven G. J. A., and Bijl H. Uncertainty quantification applied to the $k-\epsilon$ model of turbulence using the probabilistic collocation method. AIAA Paper 2008-2015, 2008.
- [13] S. B. Pope. *Turbulent Flows*, pages 358–385. Cambridge University Press, Cambridge, UK, first edition, 2000.
- [14] H. Raisi, U. Piomelli, and A. Pollard. Evaluation of turbulence models using direct numerical and large-eddy simulation data. *Journal of Fluids Engineering*, 133(2), 2011.
- [15] A. Revell, G. Iaccarino, and X. Wu. Advanced rans modeling of wingtip vortex flows. In *Annual Research Briefs, Center for Turbulence Research, NASA-AMES*, pages 173–185, 2006.
- [16] S.K. Venayagamoorthy, J.R. Koseff, J.H. Ferziger, and L.H. Shih. Testing of rans turbulence models for stratified flows based on dns data. In *Annual Research Briefs, Center for Turbulence Research, NASA-AMES*, pages 127–138, 2003.
- [17] D.C. Wilcox. *Turbulence Modeling for CFD*, pages 84–87. Griffin Printing, Glendale, CA, first edition, 1993.
- [18] M. E. Young and A. Ooi. Comparative assessment of les and urans for flow over a cylinder at a Reynolds number of 3900. In *Proceedings of the 16th Australasian Fluid Mechanics Conference*, pages 1063–1071, 2007.